TALIS Nordic Model - Sensitivity analyses for other sets of countries

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Create training and test data

```
set.seed(1312)

CountrySizes <- BTGINTT4 %>%
  count(CNTRY) %>%
  mutate(Nover=n-4000) %>%
  mutate( # select all over 4000 or 10%, whichever is bigger to be test data. this limits tra
ining data size for large countries, which was suggested by a reviewer
    Ntrain=ifelse(Nover/n <= 0.1, as.integer(n*0.1) , Nover)) %>%
  select(-Nover,-n)

BTGINTT4 %>%
  count(CNTRY,IDSCHOOL) %>%
  merge(CountrySizes) %>%
  mutate(train=0) %>%
  group_by(CNTRY) %>%
  group_modify(~{
    while ( sum(.x$train) < .x$Ntrain[1] ) {
      a <- sample(1:nrow(.x),1)
      .x[a, "train"] <- .x[a,"n"]
    }
    .x
  } ) %>%
  as.data.frame %>%
  filter(train>0) %>%
  with( paste.(CNTRY,IDSCHOOL) ) ->
  TestSchools

Test <- BTGINTT4 %>% filter(paste.(CNTRY,IDSCHOOL) %in% TestSchools )
BTGINTT4 <- BTGINTT4 %>% filter(!( paste.(CNTRY,IDSCHOOL) %in% TestSchools ) )

BTGINTT4$Nordic  |> table()
```

```
##
##      0      1
## 148616  10215
```

```
BTGINTT4$English |> table()
```

```
##
##      0      1
## 151619   7212
```

```
BTGINTT4$Random  |> table()
```

```
##
##      0      1
## 147489  11342
```

```
BTGINTT4$Random2 |> table()
```

```
##
##      0      1
## 148483  10348
```

```
BTGINTT4$Asian   |> table()
```

```
##
##      0      1
## 146175  12656
```

Run Models

```
X = BTGINTT4 %>% select(T4EXINF,T4COLES,T4SELF,T4SEMUL,T4GROMST,T4SESEN,T4COMSEL,T4SEDRT,T4BL
FDRT,T4ADVDRT,T4CLAIN,T4COGAC,T4CLASM,T4PDGENP,T4EMPS,T4CLSDIS,T4ADLE,T4PBLE,T4AUTCH,T4FULFI
L,T4OPPART,T4SESDEV,T4TLEAD,T4STUD,T4CLIBUL,T4STUHAR,T4CHFAT,T4MOPU,T4MOSU,T4WELS,T4WLOADT,T4
STBEH,T4JSENVT,T4JSPROT,T4VALP,T4JOYTCH,UseDigRes,Leadership) %>%
  as.matrix
Xtest = Test %>% select(T4EXINF,T4COLES,T4SELF,T4SEMUL,T4GROMST,T4SESEN,T4COMSEL,T4SEDRT,T4BL
FDRT,T4ADVDRT,T4CLAIN,T4COGAC,T4CLASM,T4PDGENP,T4EMPS,T4CLSDIS,T4ADLE,T4PBLE,T4AUTCH,T4FULFI
L,T4OPPART,T4SESDEV,T4TLEAD,T4STUD,T4CLIBUL,T4STUHAR,T4CHFAT,T4MOPU,T4MOSU,T4WELS,T4WLOADT,T4
STBEH,T4JSENVT,T4JSPROT,T4VALP,T4JOYTCH,UseDigRes,Leadership) %>%
  as.matrix

modelRand <- bart(
  x.train = as.matrix(X),
  y.train = factor(BTGINTT4$Random),
  x.test = Xtest,
  ntree = 50, power=2, base=0.99, k=1.5,
  nskip = 30000, ndpost = 20000, keepevery = 75,
  nchain = 12, nthread = 12,
  seed=561,
  binaryOffset=qnorm(0.071),
  keeptrees = TRUE
)

save(modelRand,file="ModelRand.Rdata")


modelEng <- bart(
  x.train = as.matrix(X),
  y.train = factor(BTGINTT4$English),
  x.test = Xtest,
  ntree = 50, power=2, base=0.99, k=1.5,
  nskip = 30000, ndpost = 20000, keepevery = 75,
  nchain = 12, nthread = 12,
  seed=561,
  keeptrees = TRUE
)

save(modelEng,file="modelEng.Rdata")


modelAsian <- bart(
  x.train = as.matrix(X),
  y.train = factor(BTGINTT4$Asian),
  x.test = Xtest,
  ntree = 50, power=2, base=0.99, k=1.5,
  nskip = 30000, ndpost = 20000, keepevery = 75,
  nchain = 12, nthread = 12,
  seed=561,
  keeptrees = TRUE
)

save(modelAsian,file="modelAsian.Rdata")


modelIsl <- bart(
```

```
  x.train = as.matrix(X),
  y.train = factor(BTGINTT4$Nordic),
  x.test = Xtest,
  ntree = 50, power=2, base=0.99, k=1.5,
  nskip = 30000, ndpost = 20000, keepevery = 75,
  nchain = 12, nthread = 12,
  seed=561,
  keeptrees = TRUE
)

save(modelIsl,file="modelIsl.Rdata")


modelRand2 <- bart(
  x.train = as.matrix(X),
  y.train = factor(BTGINTT4$Random2),
  x.test = Xtest,
  ntree = 50, power=2, base=0.99, k=1.5,
  nskip = 30000, ndpost = 20000, keepevery = 75,
  nchain = 12, nthread = 12,
  seed=561,
  binaryOffset=qnorm(0.065),
  keeptrees = TRUE
)

save(modelRand2,file="ModelRand2.Rdata")
```

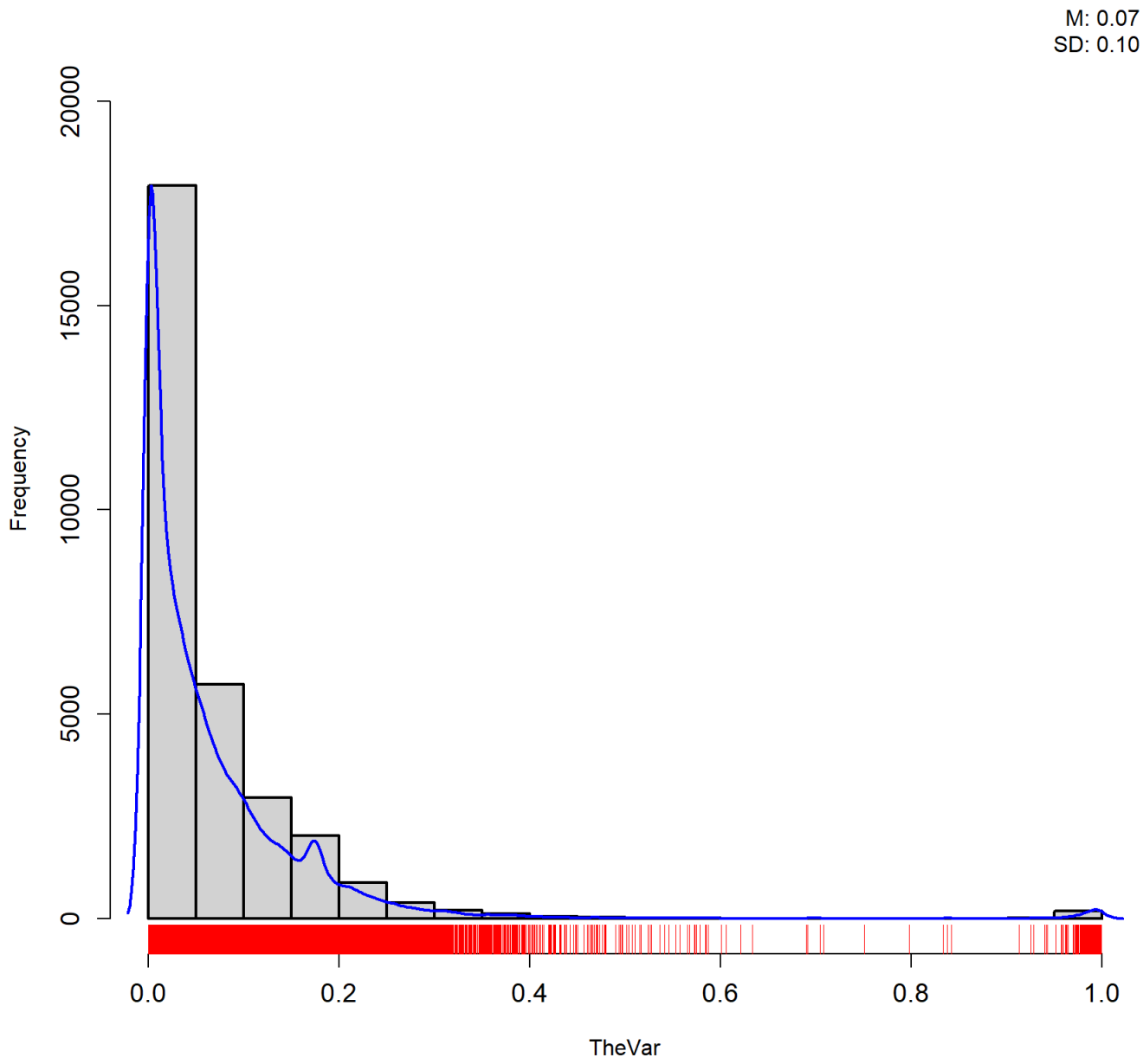
Random country Models

```
load(file="ModelRand.Rdata")
```

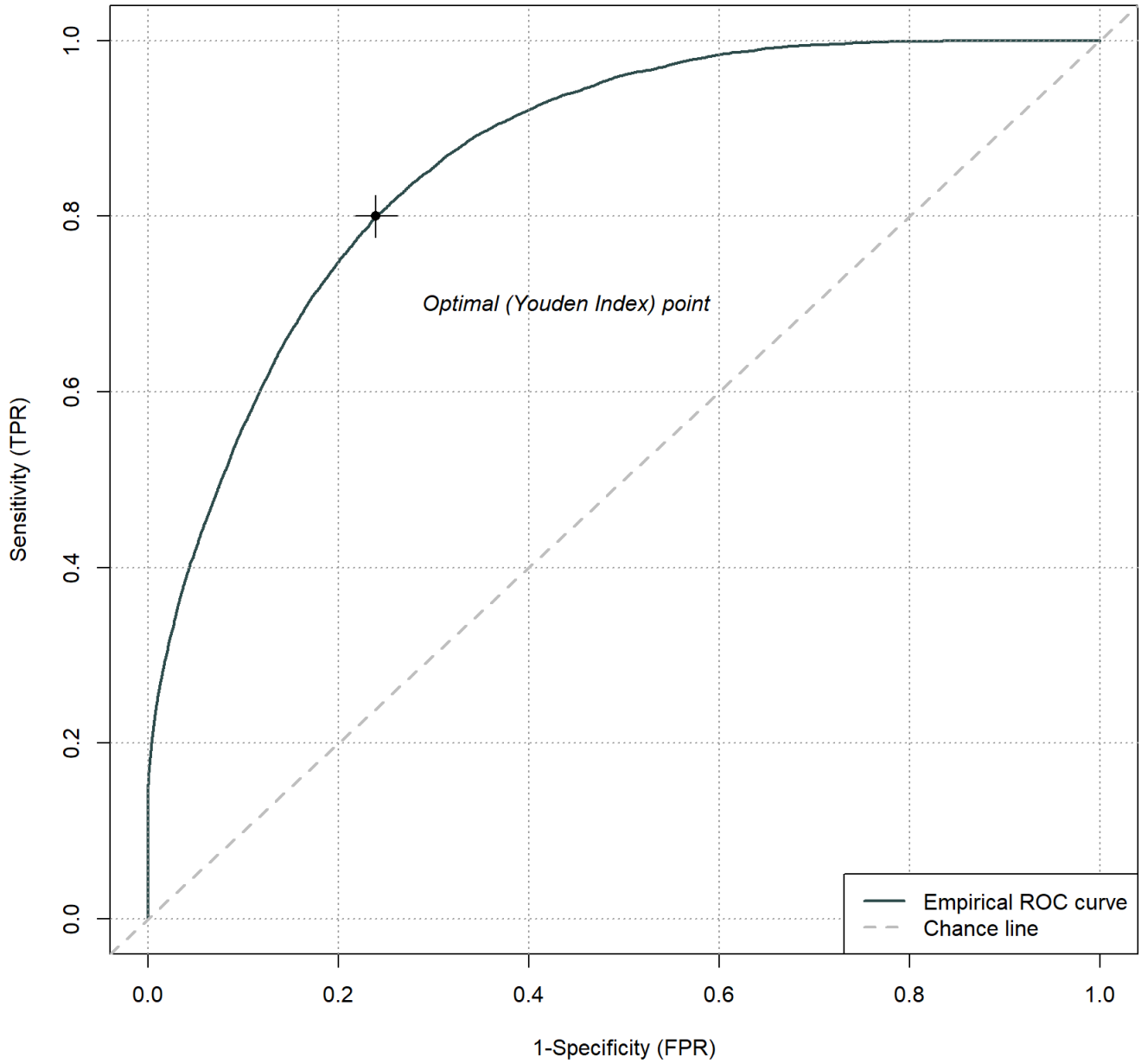
check convergence

```
hist.rug(apply(modelRand$yhat.test |> pnorm(),2,mean),main="Distribution of prediction in tes
t data")
```

```
##    vars     n mean  sd median trimmed  mad min max range skew kurtosis se
## X1    1 30515 0.07 0.1   0.03    0.05 0.05   0   1     1 4.99    37.66  0
```




```
YoudenCutoff <- plot(rocit(score=apply(modelRand$yhat.train |> pnorm(),2,mean) ,class=BTGINTT
4$Random,negref =0))
```




```
macro_predictions <- split( modelRand$yhat.train |> pnorm() %>% apply(c(1), mean) %>% t ,
                           rep(1:modelRand$call$nchain , each=modelRand$call$ndpost/modelRan
d$call$keepevery) )

mcmc_chains <- lapply(1:length(macro_predictions), function(i) {
  mcmc(macro_predictions[[i]])
}) %>%  as.mcmc.list

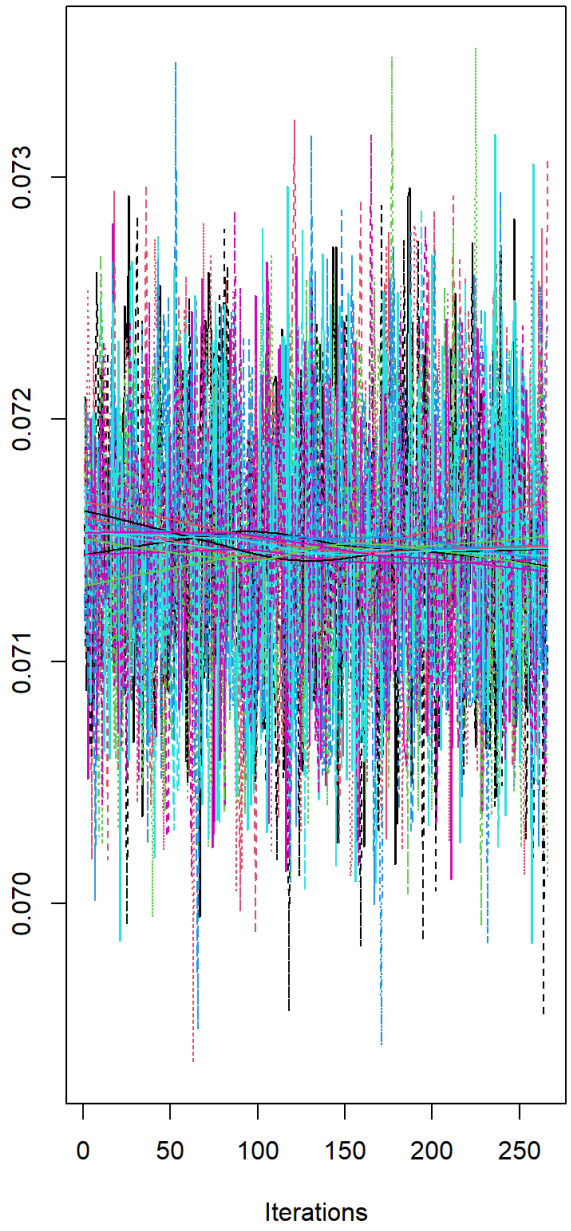
gelman.diag(mcmc_chains)
```

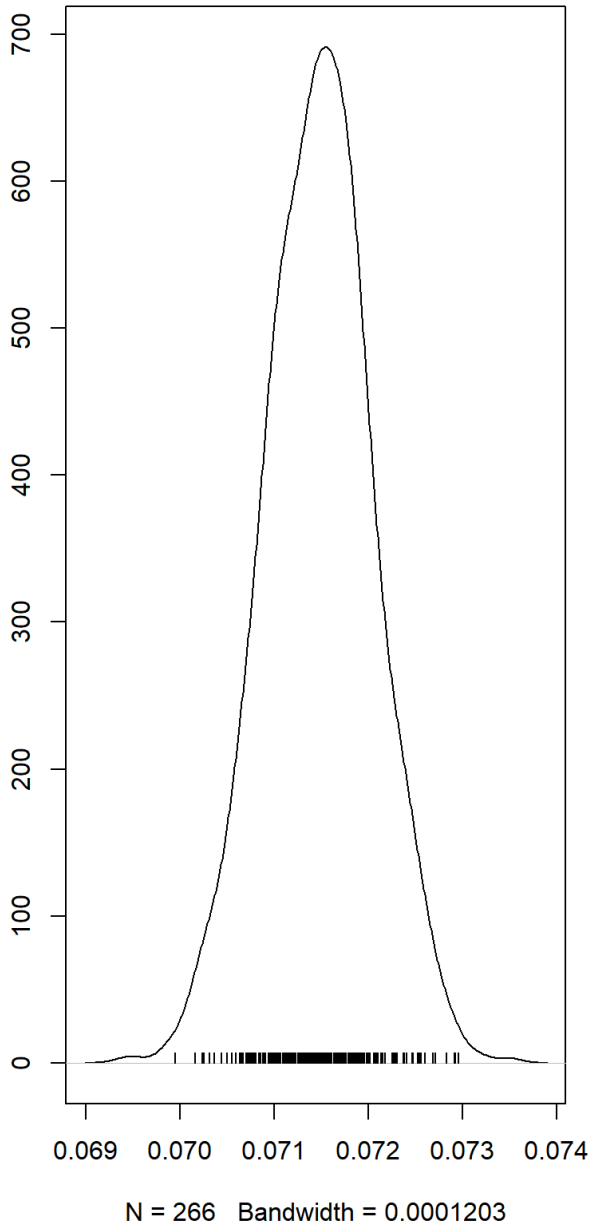
```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## [1,]          1          1
```

```
print(coda::effectiveSize(mcmc_chains))
```

```
##     var1
## 3370.306
```

```
plot(mcmc_chains)
```

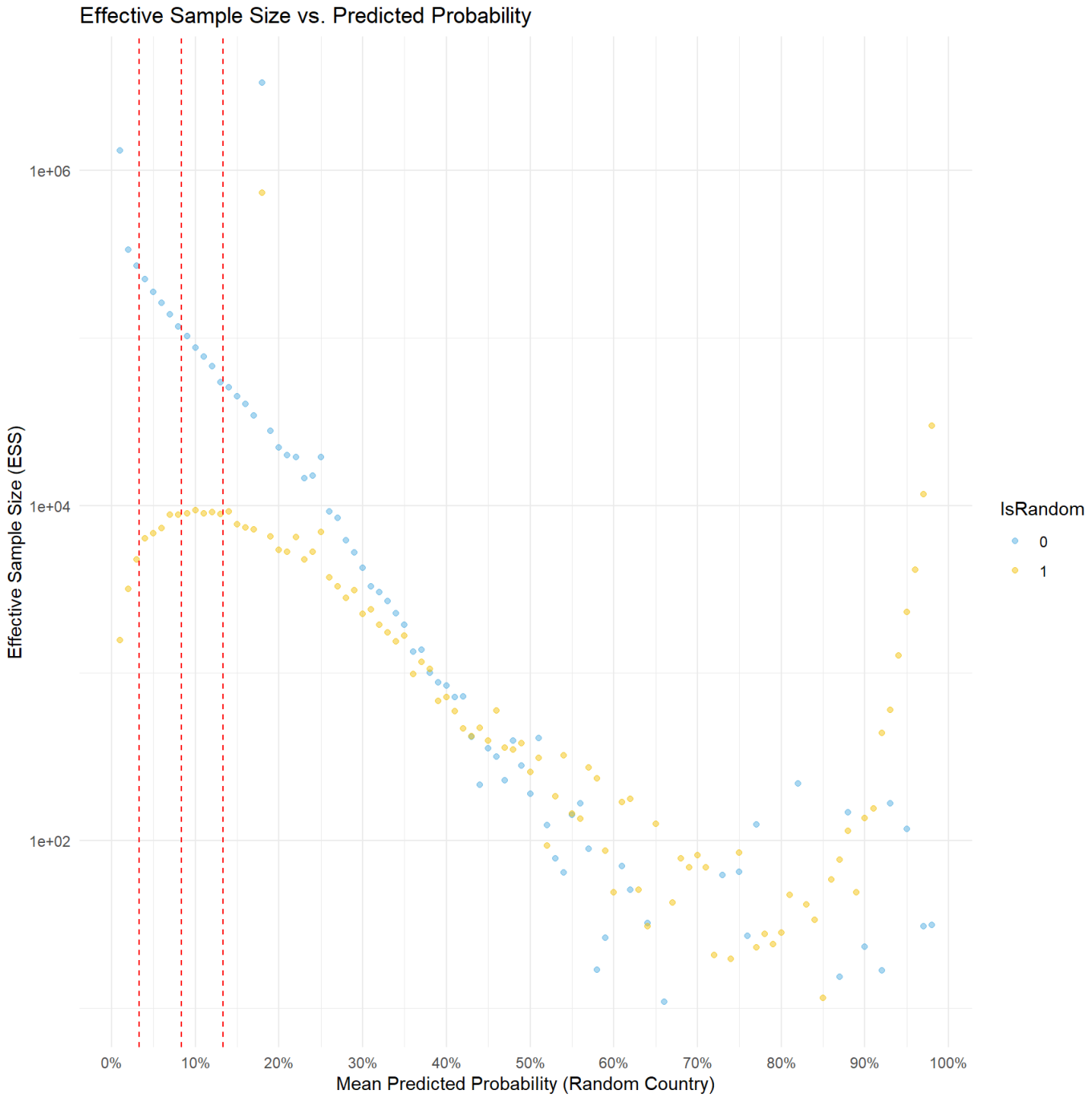




```
pred_prob_matrix <- modelRand$yhat.train |> pnorm()
# Calculate ESS for each column (teacher) by looking across samples for each case
ess_values <- apply(pred_prob_matrix, 2, effectiveSize)
# Calculate the mean probability for each teacher
mean_probs <- colMeans(pred_prob_matrix)
# Create a data frame combining Teacher ID, Mean Probability, and ESS
teacher_diagnostics <- data.frame(
  IsRandom=modelRand$fit$data@y,
  teacher_id = 1:length(mean_probs),
  mean_prob = mean_probs,
  ess = ess_values  )

cutoff_zone <- teacher_diagnostics[teacher_diagnostics$mean_prob >= YoudenCutoff$optimal[4]-
0.05 &
                                   teacher_diagnostics$mean_prob <= YoudenCutoff$optimal[4]+
0.05, ]

teacher_diagnostics |>
  mutate(mean_prob=cut(mean_prob,seq(0,1,0.01),include.lowest=T) |> factor() |> as.integer()
,
         mean_prob=mean_prob/100,
         IsRandom=factor(IsRandom) ) |>
  group_by(mean_prob,IsRandom) |>
  summarize(ess=sum(ess,na.rm=T) , .groups="drop") |>
  ggplot(aes(x = mean_prob, y = ess,color=IsRandom)) +
  geom_point(alpha = 0.5) +
  geom_vline(xintercept = c(max(YoudenCutoff$optimal[4]-0.05,0),
                            YoudenCutoff$optimal[4],
                            min(YoudenCutoff$optimal[4]+0.05,1)),
             linetype = "dashed", color = "red") +
  labs(title = "Effective Sample Size vs. Predicted Probability",
       x = "Mean Predicted Probability (Random Country)",
       y = "Effective Sample Size (ESS)") +
  theme_minimal()+
  scale_x_continuous(breaks=seq(0,1,0.1),label=scales::percent)+
  scale_y_log10()
```

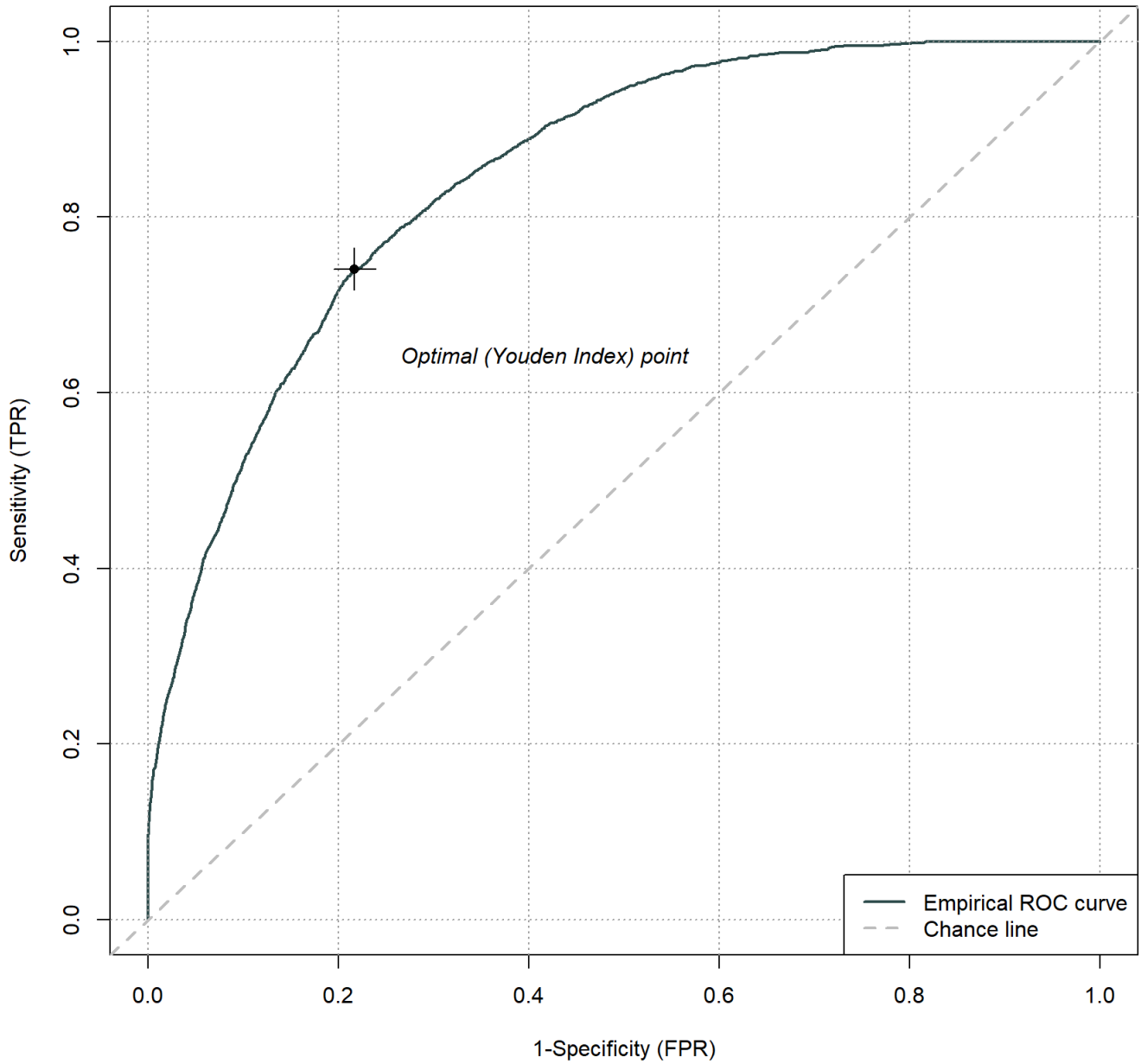


Results

```
str(YoudenCutoff)
```

```
## List of 6
##  $ method                    : chr "empirical"
##  $ AUC                       : num 0.867
##  $ Cutoff                    : num [1:156144] Inf 1 1 1 1 ...
##  $ TPR/Sensitivity/Recall    : num [1:156144] 0.00 8.82e-05 1.76e-04 2.65e-04 3.53e-04 ...
##  $ FPR/1-Specificity         : num [1:156144] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.5608 0.2392 0.8 0.0832
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

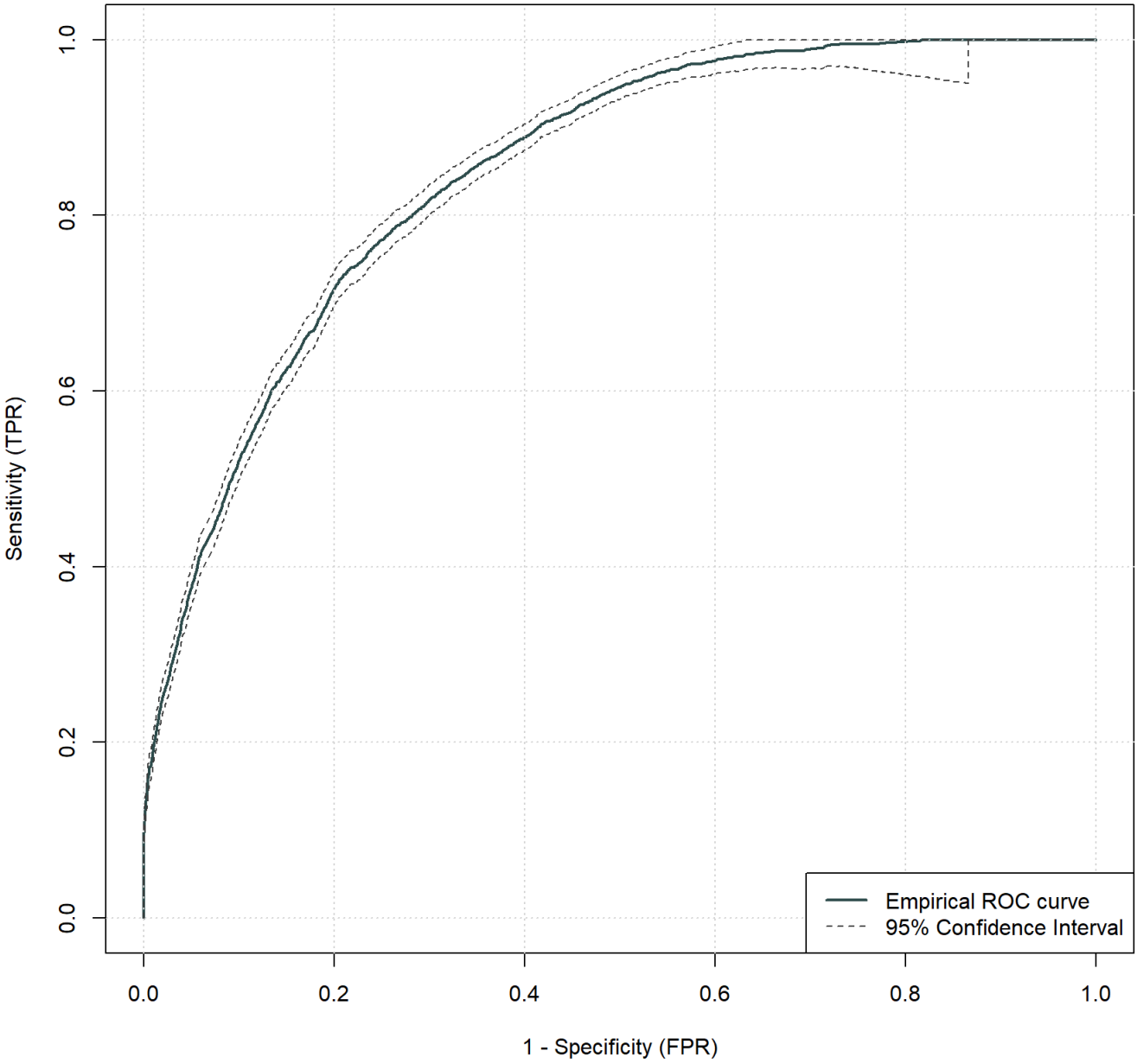
```
ROCit_obj_Random       <- rocit(score=apply(modelRand$yhat.test |> pnorm(),2,mean) ,
                                class=Test$Random,negref =0)
str( ROCIT_Random <- plot(ROCit_obj_Random) )
```




```
## List of 6
##  $ method                   : chr "empirical"
##  $ AUC                      : num 0.847
##  $ Cutoff                   : num [1:30079] Inf 1 1 1 1 ...
##  $ TPR/Sensitivity/Recall   : num [1:30079] 0 0.000496 0.000991 0.001487 0.001982 ...
##  $ FPR/1-Specificity        : num [1:30079] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.5241 0.2167 0.7408 0.0907
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

```
ciROC(ROCit_obj_Random, method = "binomial") %>% plot(legend = TRUE)
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm): collapsing to uniq
ue 'x' values
```




```
ciAUC(ROCit_obj_Random, delong = TRUE)
```

```
##
##    estimated AUC : 0.846749564826482
##    AUC estimation method : empirical
##
##    CI of AUC, delong method of variance used
##    confidence level = 95%
##    lower = 0.838988006372546     upper = 0.854511123280418
```

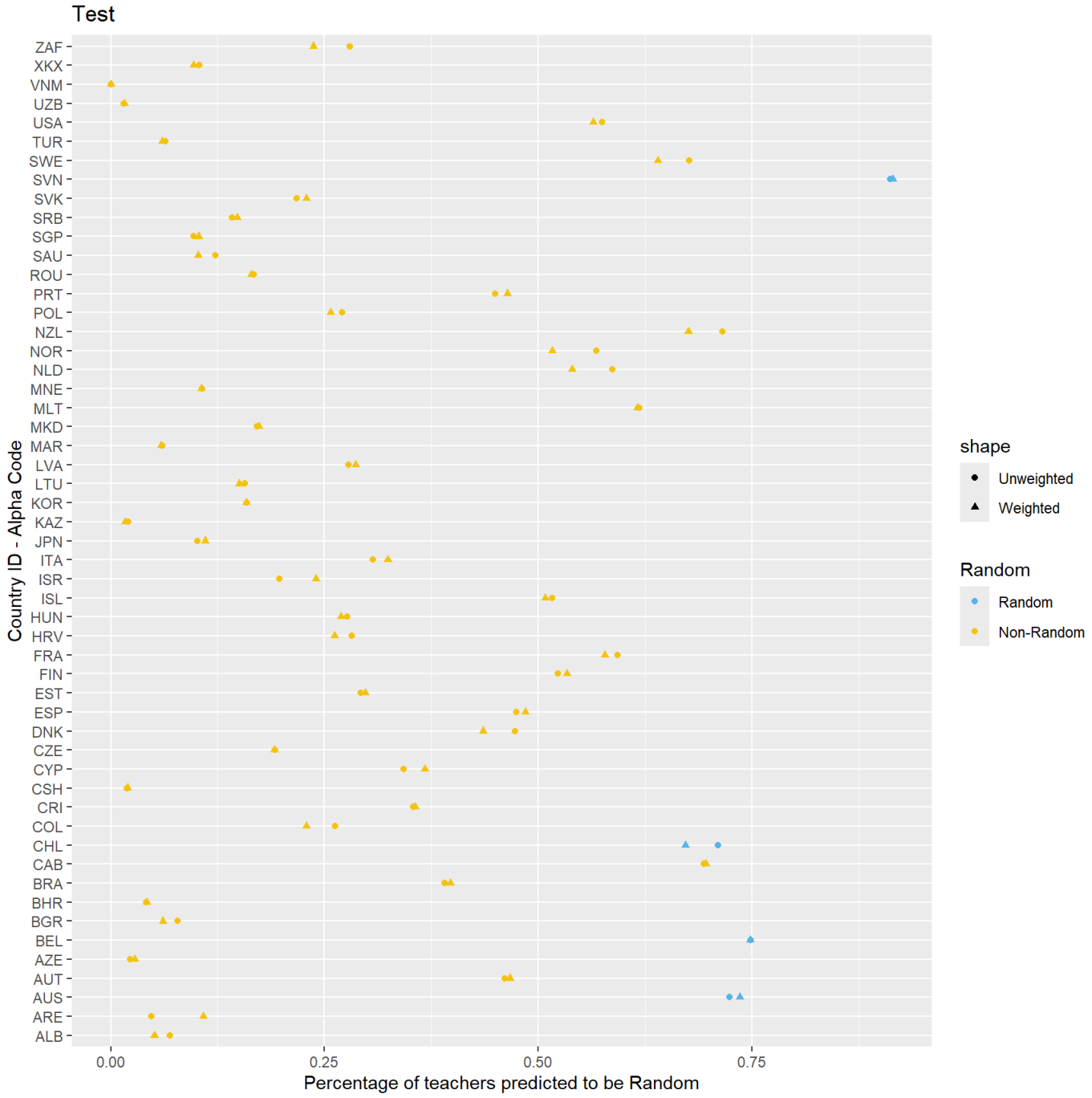
```
save(ROCIT_Random,ROCit_obj_Random,YoudenCutoff,file="ROCIT_Random_Test.rData")
```

Tables for classification

```
Test %>%
  transmute(CNTRY,predicted=apply(modelRand$yhat.test |> pnorm(),2,mean) >= YoudenCutoff$opti
mal[4]) %>%
  table
```

```
##      predicted
## CNTRY FALSE TRUE
##   ALB   376   28
##   ARE   418   21
##   AUS    87  228
##   AUT   243  208
##   AZE   425   10
##   BEL   298  886
##   BGR   355   30
##   BHR   300   13
##   BRA   189  121
##   CAB    59  134
##   CHL    68  167
##   COL   233   83
##   CRI   199  109
##   CSH   405    8
##   CYP   123   64
##   CZE  1921  457
##   DNK   135  121
##   ESP  2681 2424
##   EST   242  100
##   FIN   214  235
##   FRA   119  173
##   HRV   227   89
##   HUN   293  112
##   ISL    73   78
##   ISR   268   66
##   ITA   271  120
##   JPN   329   37
##   KAZ  3299   68
##   KOR   264   50
##   LTU   365   68
##   LVA   296  114
##   MAR  1988  128
##   MKD   335   69
##   MLT    97  157
##   MNE   126   15
##   NLD    55   78
##   NOR    67   88
##   NZL    37   93
##   POL   277  103
##   PRT   203  166
##   ROU   329   66
##   SAU   293   41
##   SGP   307   33
##   SRB   356   59
##   SVK   334   93
##   SVN    25  259
##   SWE    99  207
##   TUR   893   61
##   USA    85  115
##   UZB   452    7
##   VNM   450    0
##   XKX   295   34
##   ZAF   247   96
```

```
Test %>%
  mutate(predicted = apply(modelRand$yhat.test |> pnorm(),2,mean),
         Random=factor(Random,1:0,c("Random","Non-Random")) ) %>%
  group_by(CNTRY,Random) %>%
  summarize(Unwg_pred=mean(predicted>=YoudenCutoff$optimal[4]),
            Wgt_pred =weighted.mean(predicted>=YoudenCutoff$optimal[4],unlabelled(TCHWGT)) ,
            .groups="drop") %>%
  ggplot(aes(y=CNTRY,color=Random))+
    geom_point(aes(x=Unwg_pred,shape="Unweighted"))+
    geom_point(aes(x=Wgt_pred,shape="Weighted"))+
    labs(x="Percentage of teachers predicted to be Random",title="Test")
```



remove variables to create space in RAM

```
rm(modelRand, ROCit_obj_Random,ROCIT_Random,mcmc_chains,macro_predictions)
gc()
```

```
##             used   (Mb) gc trigger    (Mb)   max used    (Mb)
## Ncells   4041141  215.9    7386084   394.5    7386084   394.5
## Vcells 526309535 4015.5 2635081462 20104.1 3427413811 26149.1
```

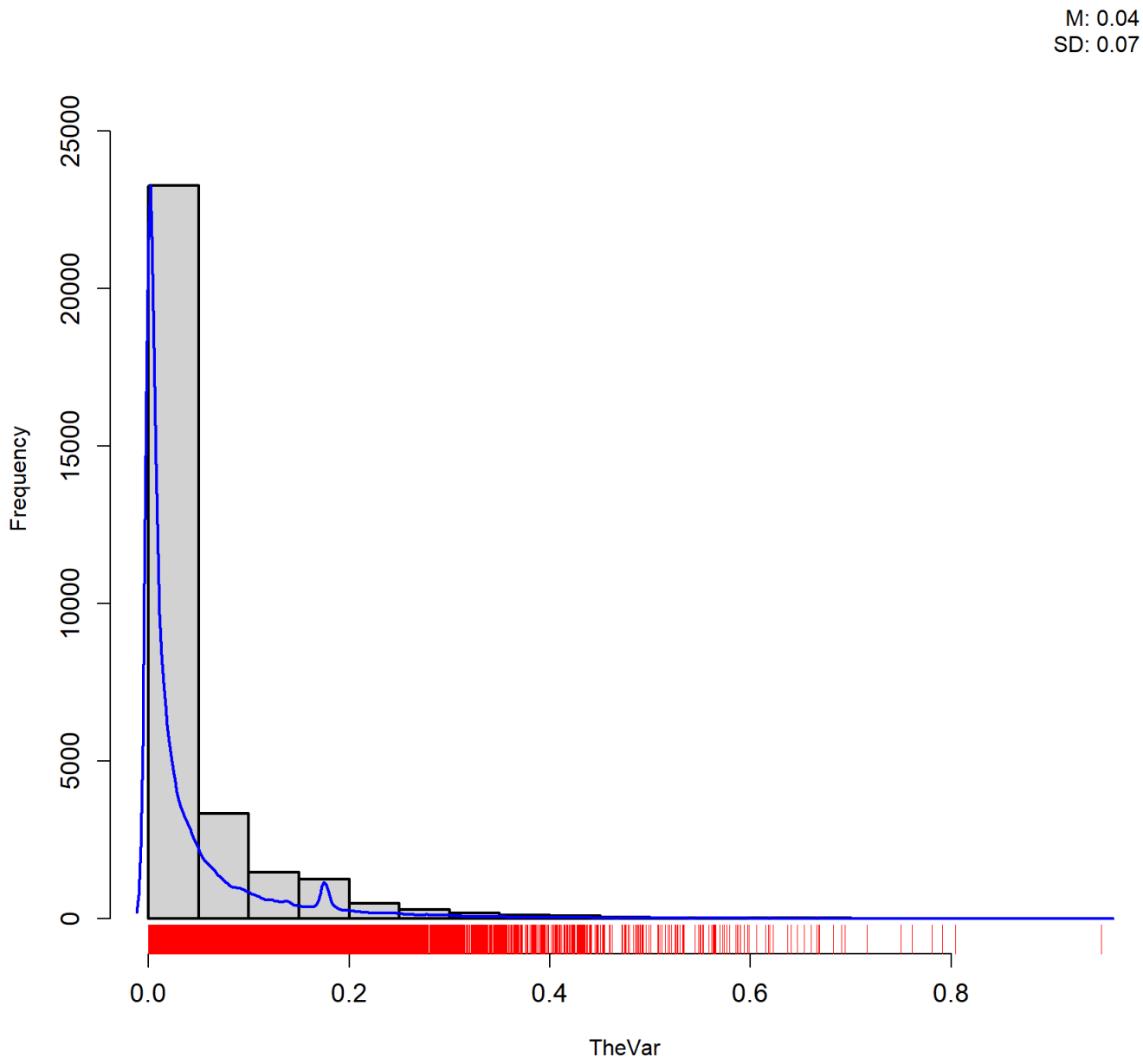
Models for English Countries

```
load(file="modelEng.Rdata")
```

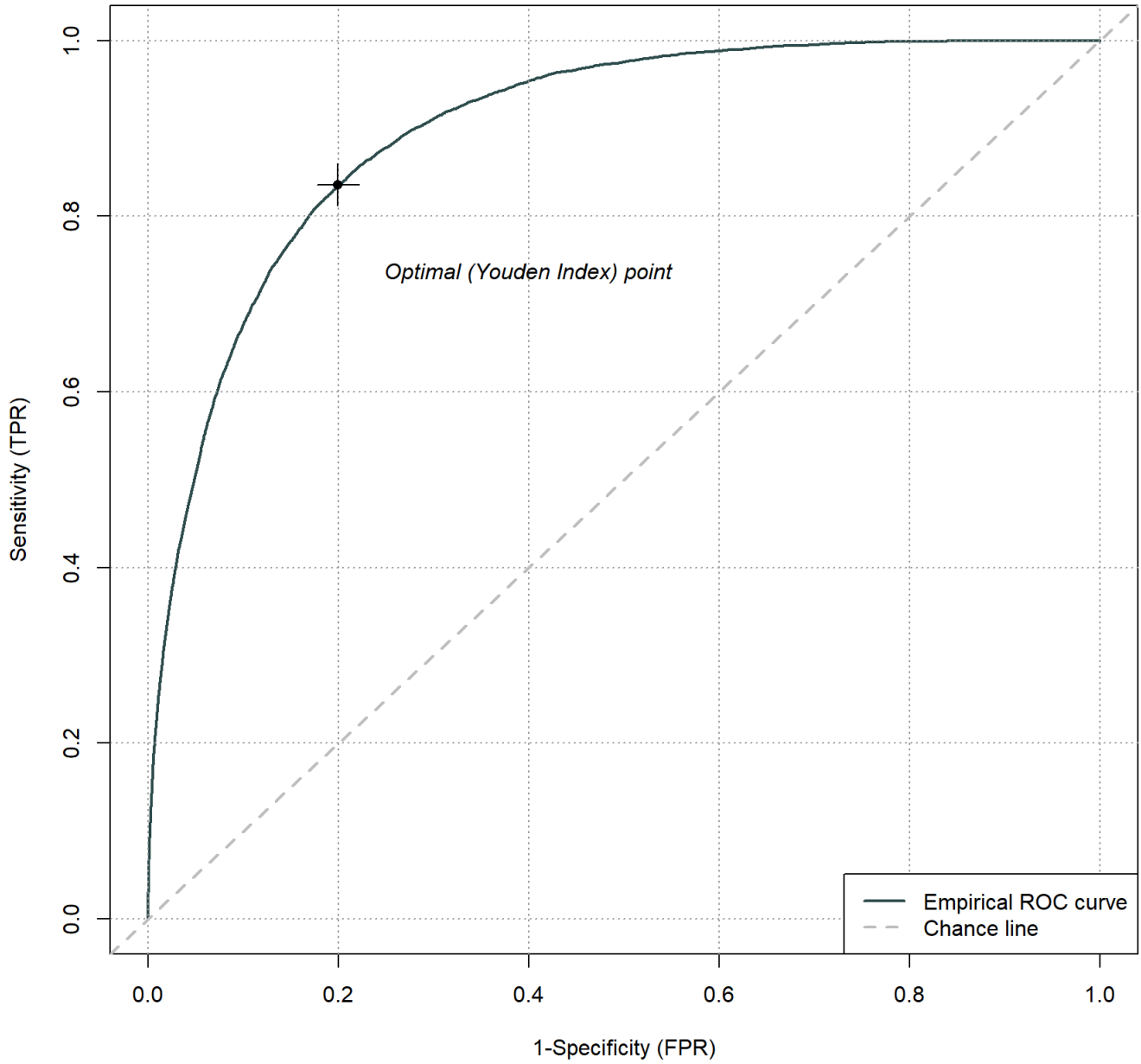
check convergence

```
hist.rug(apply(modelEng$yhat.test |> pnorm(),2,mean),main="Distribution of prediction in test
data")
```

```
##    vars     n mean   sd median trimmed  mad min  max range skew kurtosis se
## X1    1 30515 0.04 0.07   0.01    0.02 0.02   0 0.95  0.95 3.31    15.45  0
```




```
YoudenCutoff <- plot(rocit(score=apply(modelEng$yhat.train |> pnorm(),2,mean) ,
                           class=BTGINTT4$English,negref =0))
```




```
macro_predictions <- split( modelEng$yhat.train |> pnorm() |> apply(c(1), mean) %>% t ,
                           rep(1:modelEng$call$nchain , each=modelEng$call$ndpost/modelEng$c
all$keepevery) )

mcmc_chains <- lapply(1:length(macro_predictions), function(i) {
  mcmc(macro_predictions[[i]])
}) %>%  as.mcmc.list

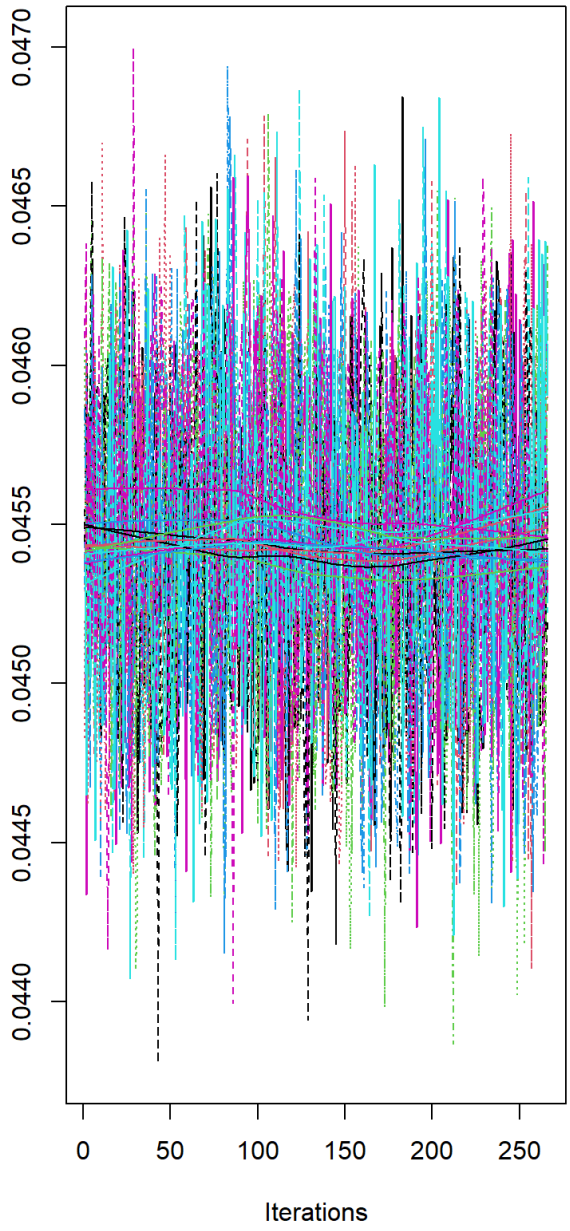
gelman.diag(mcmc_chains)
```

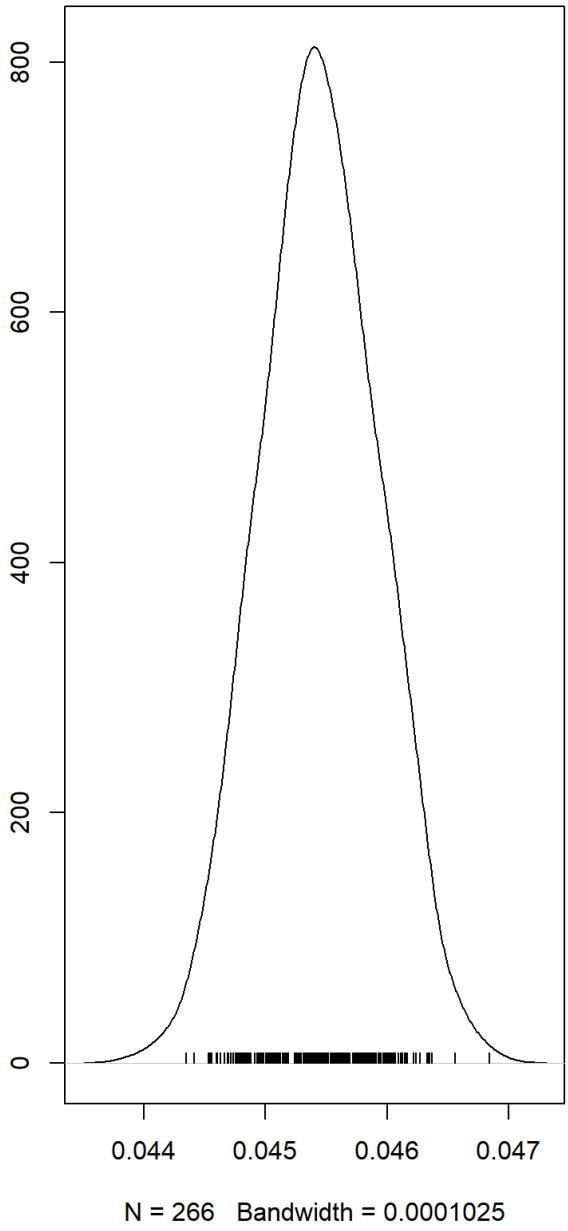
```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## [1,]          1       1.01
```

```
print(coda::effectiveSize(mcmc_chains))
```

```
##     var1
## 3143.739
```

```
plot(mcmc_chains)
```

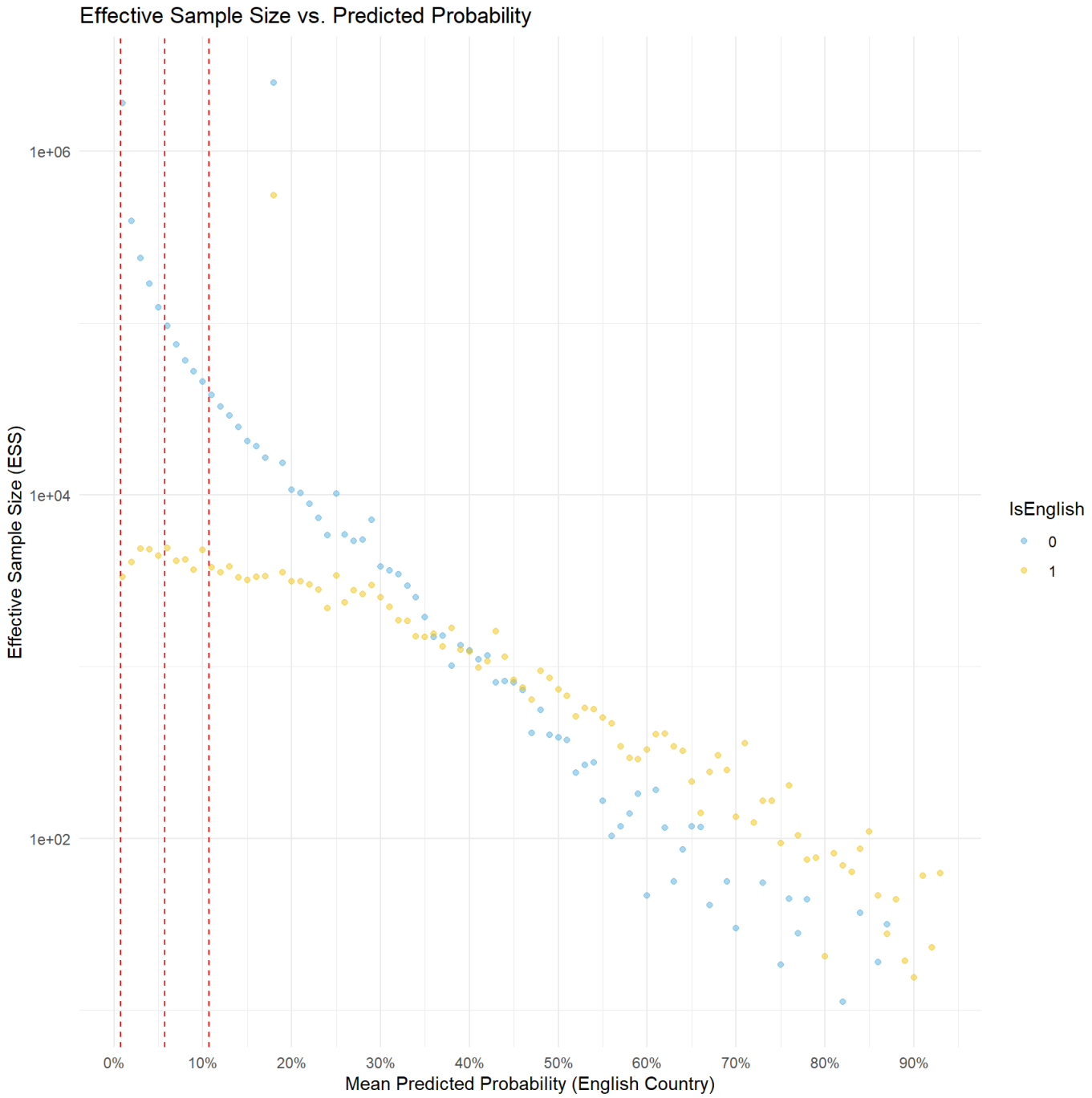




```
pred_prob_matrix <- modelEng$yhat.train |> pnorm()
# Calculate ESS for each column (teacher) by looking across samples for each case
ess_values <- apply(pred_prob_matrix, 2, effectiveSize)
# Calculate the mean probability for each teacher
mean_probs <- colMeans(pred_prob_matrix)
# Create a data frame combining Teacher ID, Mean Probability, and ESS
teacher_diagnostics <- data.frame(
  IsEnglish=modelEng$fit$data@y,
  teacher_id = 1:length(mean_probs),
  mean_prob = mean_probs,
  ess = ess_values  )

cutoff_zone <- teacher_diagnostics[teacher_diagnostics$mean_prob >= YoudenCutoff$optimal[4]-
0.05 &
                                   teacher_diagnostics$mean_prob <= YoudenCutoff$optimal[4]+
0.05, ]

teacher_diagnostics |>
  mutate(mean_prob=cut(mean_prob,seq(0,1,0.01),include.lowest=T) |> factor() |> as.integer()
,
         mean_prob=mean_prob/100,
         IsEnglish=factor(IsEnglish) ) |>
  group_by(mean_prob,IsEnglish) |>
  summarize(ess=sum(ess,na.rm=T) , .groups="drop") |>
  ggplot(aes(x = mean_prob, y = ess,color=IsEnglish)) +
  geom_point(alpha = 0.5) +
  geom_vline(xintercept = c(max(YoudenCutoff$optimal[4]-0.05,0),
                            YoudenCutoff$optimal[4],
                            min(YoudenCutoff$optimal[4]+0.05,1)),
             linetype = "dashed", color = "red") +
  labs(title = "Effective Sample Size vs. Predicted Probability",
       x = "Mean Predicted Probability (English Country)",
       y = "Effective Sample Size (ESS)") +
  theme_minimal()+
  scale_x_continuous(breaks=seq(0,1,0.1),label=scales::percent)+
  scale_y_log10()
```

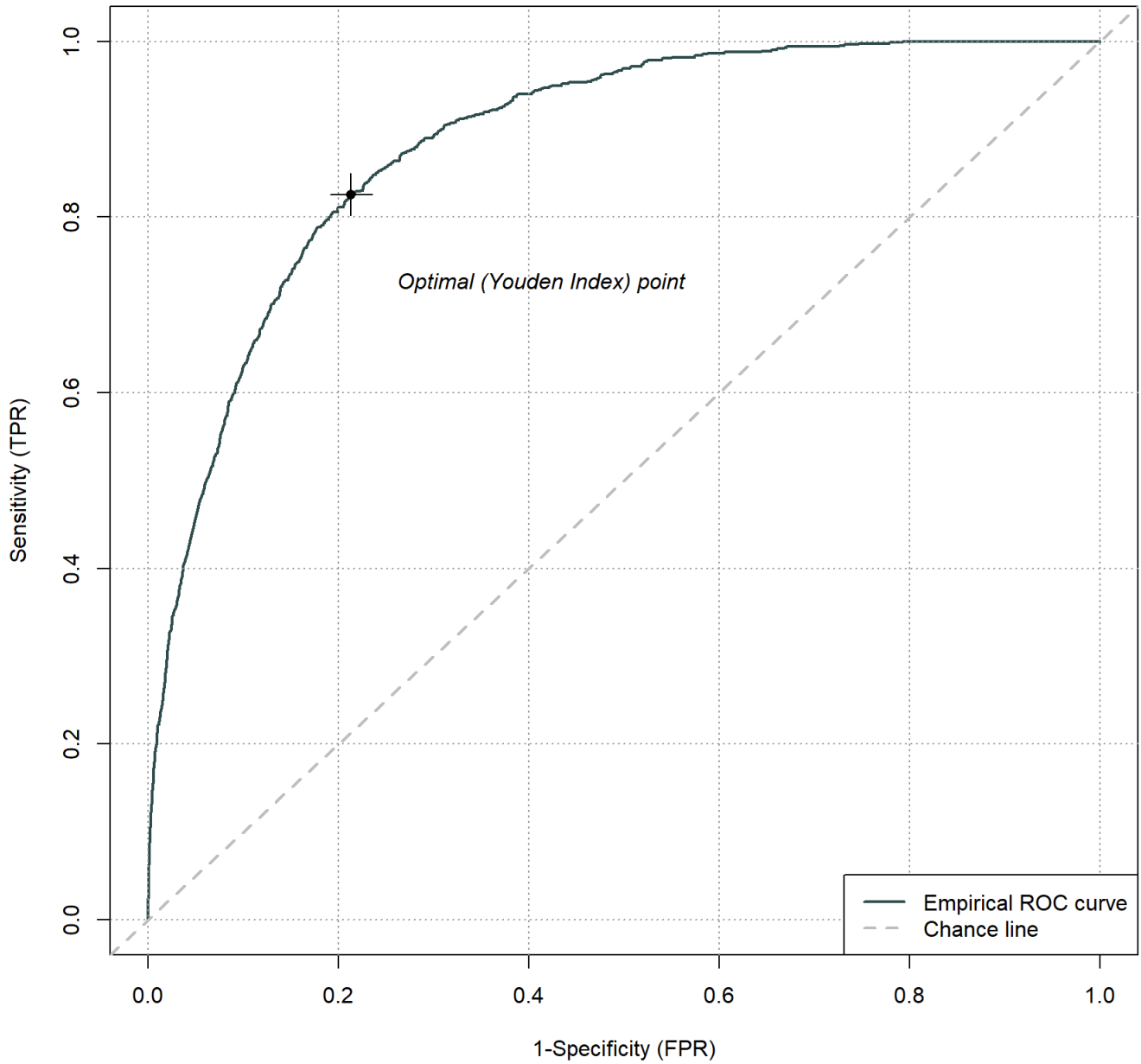


Results English

```
str( YoudenCutoff )
```

```
## List of 6
##  $ method                    : chr "empirical"
##  $ AUC                       : num 0.899
##  $ Cutoff                    : num [1:156144] Inf 0.952 0.951 0.937 0.919 ...
##  $ TPR/Sensitivity/Recall    : num [1:156144] 0 0.000139 0.000277 0.000416 0.000555 ...
##  $ FPR/1-Specificity         : num [1:156144] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.6367 0.1993 0.836 0.0573
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

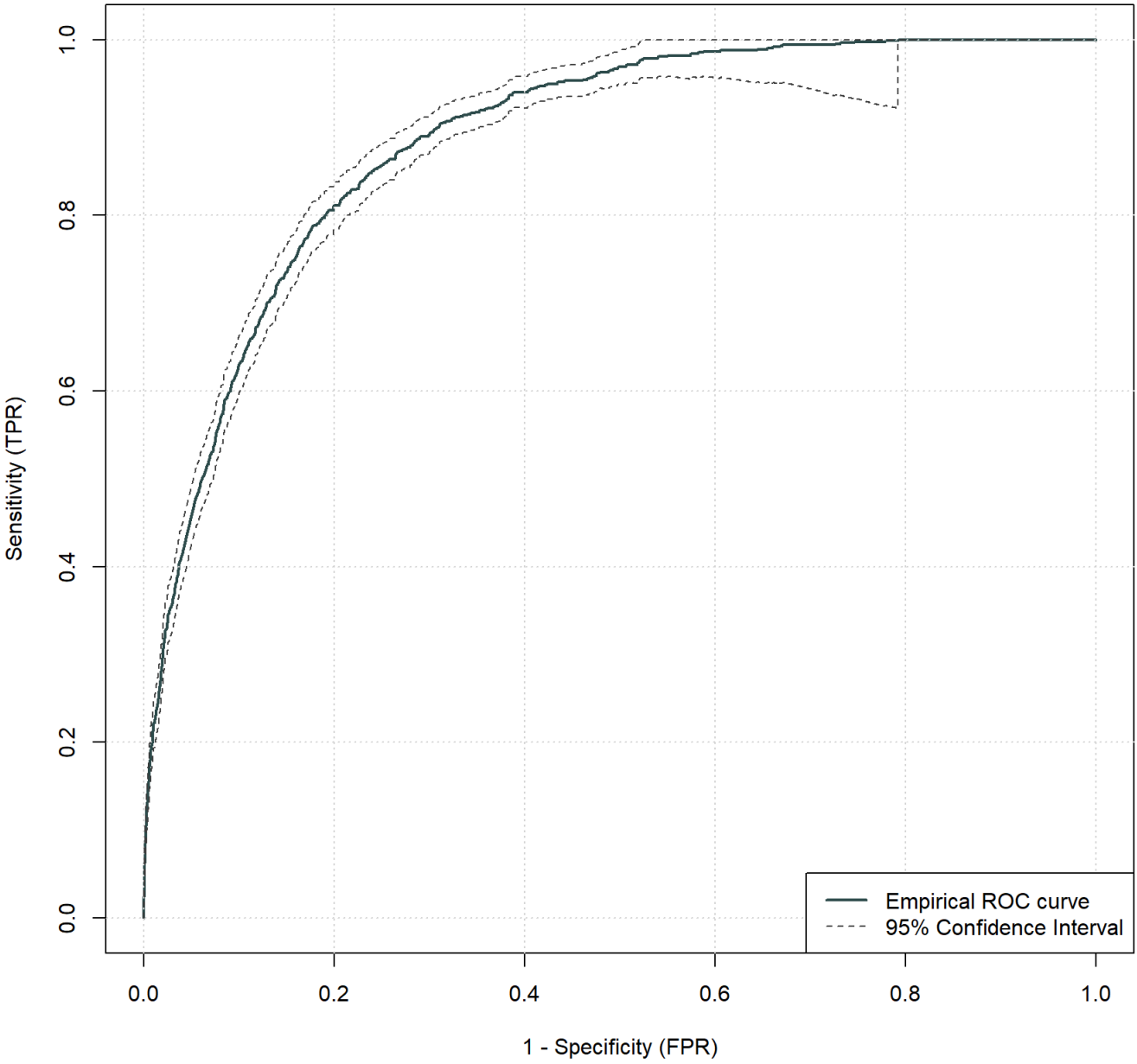
```
ROCit_obj_English      <- rocit(score=apply(modelEng$yhat.test |> pnorm(),2,mean),
                                class=Test$English,negref =0)
str( ROCIT_English <- plot(ROCit_obj_English) )
```




```
## List of 6
##  $ method                   : chr "empirical"
##  $ AUC                      : num 0.886
##  $ Cutoff                   : num [1:30079] Inf 0.95 0.804 0.791 0.781 ...
##  $ TPR/Sensitivity/Recall   : num [1:30079] 0 0.00119 0.00119 0.00239 0.00358 ...
##  $ FPR/1-Specificity        : num [1:30079] 0.00 0.00 3.37e-05 3.37e-05 3.37e-05 ...
##  $ optimal Youden Index point: Named num [1:4] 0.6126 0.2132 0.8258 0.0521
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

```
ciROC(ROCit_obj_English, method = "binomial") %>% plot(legend = TRUE)
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm): collapsing to uniq
ue 'x' values
```




```
ciAUC(ROCit_obj_English, delong = TRUE)
```

```
##
##    estimated AUC : 0.885682647772602
##    AUC estimation method : empirical
##
##    CI of AUC, delong method of variance used
##    confidence level = 95%
##    lower = 0.875826052209215     upper = 0.895539243335988
```

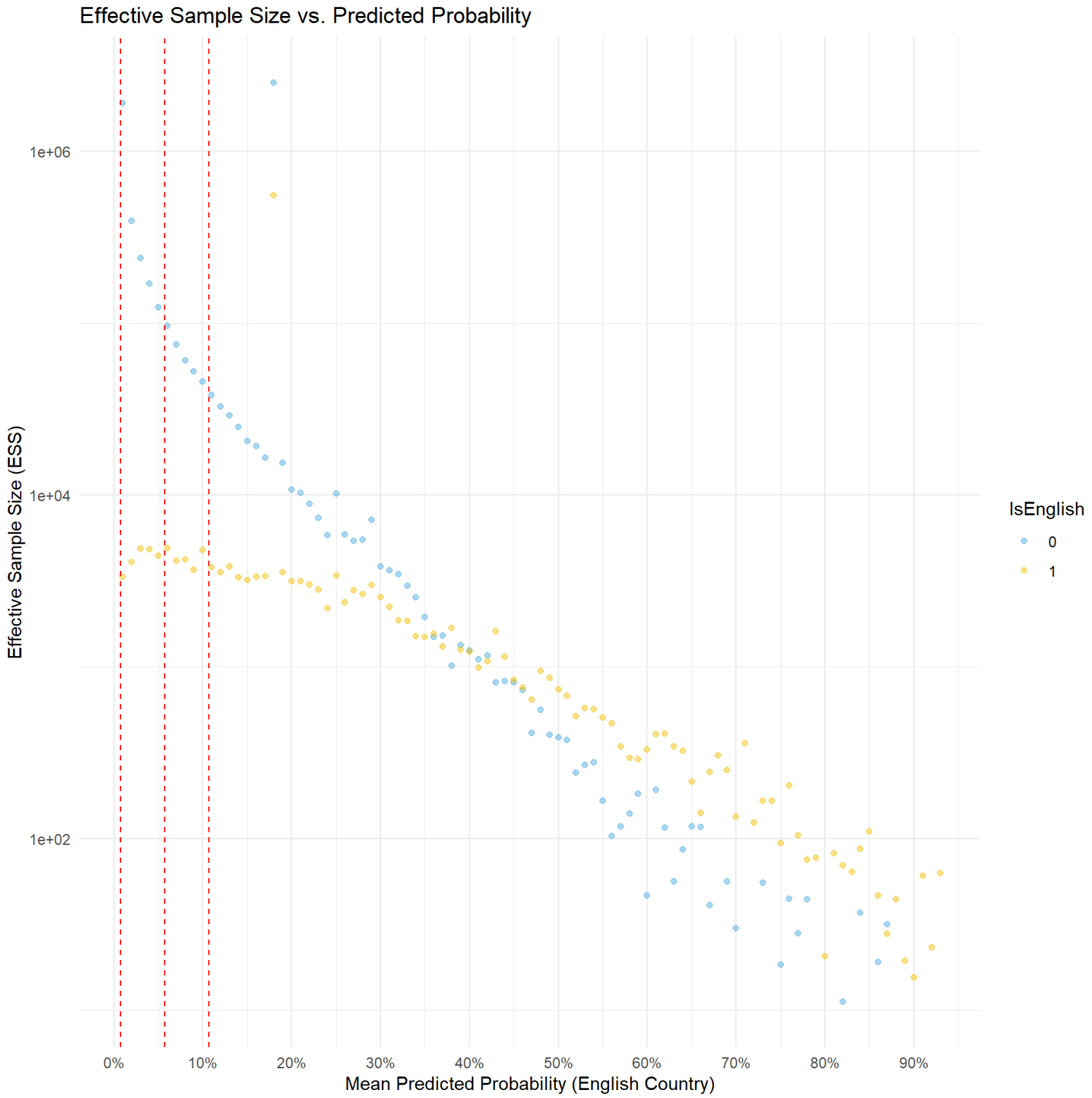
```
save(ROCIT_English,ROCit_obj_English,YoudenCutoff,file="ROCIT_English_Test.rData")



pred_prob_matrix <- modelEng$yhat.train |> pnorm()
# Calculate ESS for each column (teacher) by looking across samples for each case
ess_values <- apply(pred_prob_matrix, 2, effectiveSize)
# Calculate the mean probability for each teacher
mean_probs <- colMeans(pred_prob_matrix)
# Create a data frame combining Teacher ID, Mean Probability, and ESS
teacher_diagnostics <- data.frame(
  IsEnglish=modelEng$fit$data@y,
  teacher_id = 1:length(mean_probs),
  mean_prob = mean_probs,
  ess = ess_values  )

cutoff_zone <- teacher_diagnostics[teacher_diagnostics$mean_prob >= YoudenCutoff$optimal[4]-
0.05 &
                                   teacher_diagnostics$mean_prob <= YoudenCutoff$optimal[4]+
0.05, ]

teacher_diagnostics |>
  mutate(mean_prob=cut(mean_prob,seq(0,1,0.01),include.lowest=T) |> factor() |> as.integer()
,
         mean_prob=mean_prob/100,
         IsEnglish=factor(IsEnglish) ) |>
  group_by(mean_prob,IsEnglish) |>
  summarize(ess=sum(ess,na.rm=T) , .groups="drop") |>
  ggplot(aes(x = mean_prob, y = ess,color=IsEnglish)) +
  geom_point(alpha = 0.5) +
  geom_vline(xintercept = c(max(YoudenCutoff$optimal[4]-0.05,0),
                            YoudenCutoff$optimal[4],
                            min(YoudenCutoff$optimal[4]+0.05,1)),
             linetype = "dashed", color = "red") +
  labs(title = "Effective Sample Size vs. Predicted Probability",
       x = "Mean Predicted Probability (English Country)",
       y = "Effective Sample Size (ESS)") +
  theme_minimal()+
  scale_x_continuous(breaks=seq(0,1,0.1),label=scales::percent)+
  scale_y_log10()
```

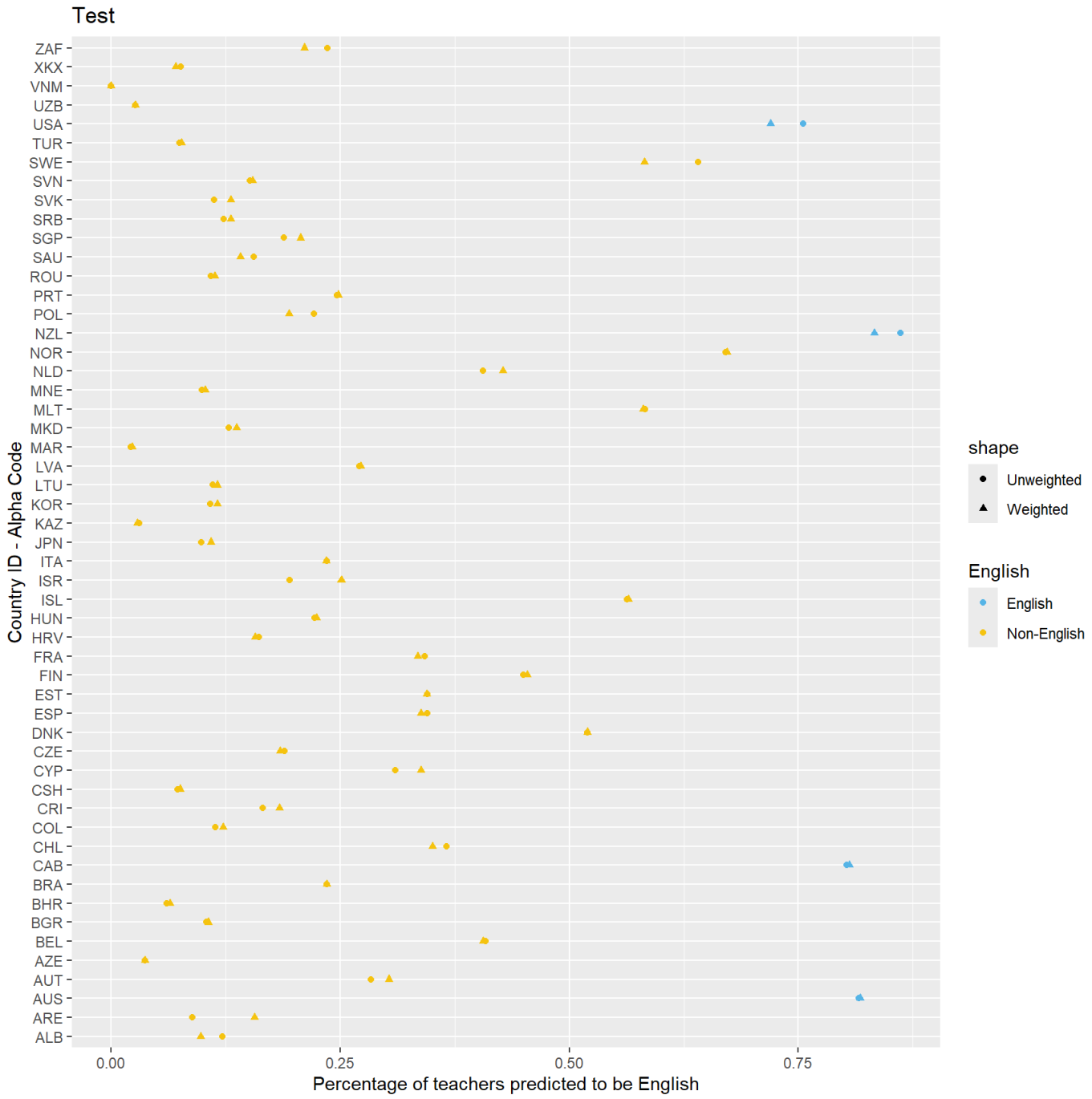


Tables for classification

```
Test %>%
  transmute(CNTRY,predicted=apply(modelEng$yhat.test |> pnorm(),2,mean)>=YoudenCutoff$optimal
[4]) %>%
  table
```

```
##      predicted
## CNTRY FALSE TRUE
##   ALB   355   49
##   ARE   400   39
##   AUS    58  257
##   AUT   323  128
##   AZE   419   16
##   BEL   700  484
##   BGR   345   40
##   BHR   294   19
##   BRA   237   73
##   CAB    38  155
##   CHL   149   86
##   COL   280   36
##   CRI   257   51
##   CSH   383   30
##   CYP   129   58
##   CZE  1928  450
##   DNK   123  133
##   ESP  3343 1762
##   EST   224  118
##   FIN   247  202
##   FRA   192  100
##   HRV   265   51
##   HUN   315   90
##   ISL    66   85
##   ISR   269   65
##   ITA   299   92
##   JPN   330   36
##   KAZ  3263  104
##   KOR   280   34
##   LTU   385   48
##   LVA   299  111
##   MAR  2071   45
##   MKD   352   52
##   MLT   106  148
##   MNE   127   14
##   NLD    79   54
##   NOR    51  104
##   NZL    18  112
##   POL   296   84
##   PRT   278   91
##   ROU   352   43
##   SAU   282   52
##   SGP   276   64
##   SRB   364   51
##   SVK   379   48
##   SVN   241   43
##   SWE   110  196
##   TUR   883   71
##   USA    49  151
##   UZB   447   12
##   VNM   450    0
##   XKX   304   25
##   ZAF   262   81
```

```
Test %>%
  mutate(predicted = apply(modelEng$yhat.test |> pnorm(),2,mean),
         English=factor(English,1:0,c("English","Non-English")) ) %>%
  group_by(CNTRY,English) %>%
  summarize(Unwg_pred=mean(predicted>=YoudenCutoff$optimal[4]),
            Wgt_pred =weighted.mean(predicted>=YoudenCutoff$optimal[4],unlabelled(TCHWGT)) ,
            .groups="drop") %>%
  ggplot(aes(y=CNTRY,color=English))+
    geom_point(aes(x=Unwg_pred,shape="Unweighted"))+
    geom_point(aes(x=Wgt_pred,shape="Weighted"))+
    labs(x="Percentage of teachers predicted to be English",title="Test")
```



remove variables to create space in RAM

```
rm(modelEng, ROCIT_English,ROCit_obj_English,mcmc_chains,macro_predictions)
gc()
```

```
##             used   (Mb) gc trigger    (Mb)   max used    (Mb)
## Ncells   4041185  215.9    7386087   394.5    7386087   394.5
## Vcells 526378271 4016.0 2529742204 19300.5 3427413811 26149.1
```

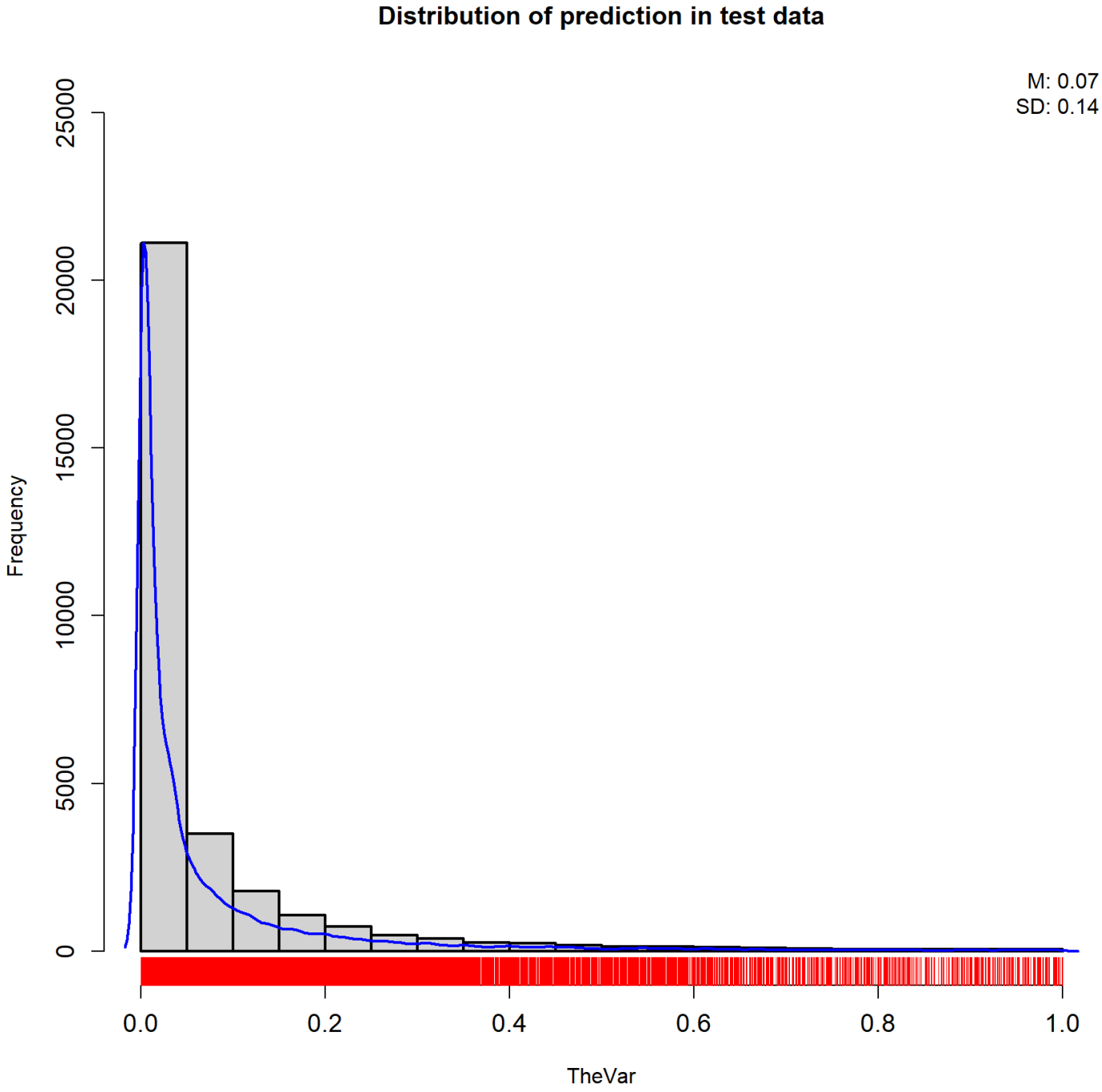
Asian Models

```
load(file="modelAsian.Rdata")
```

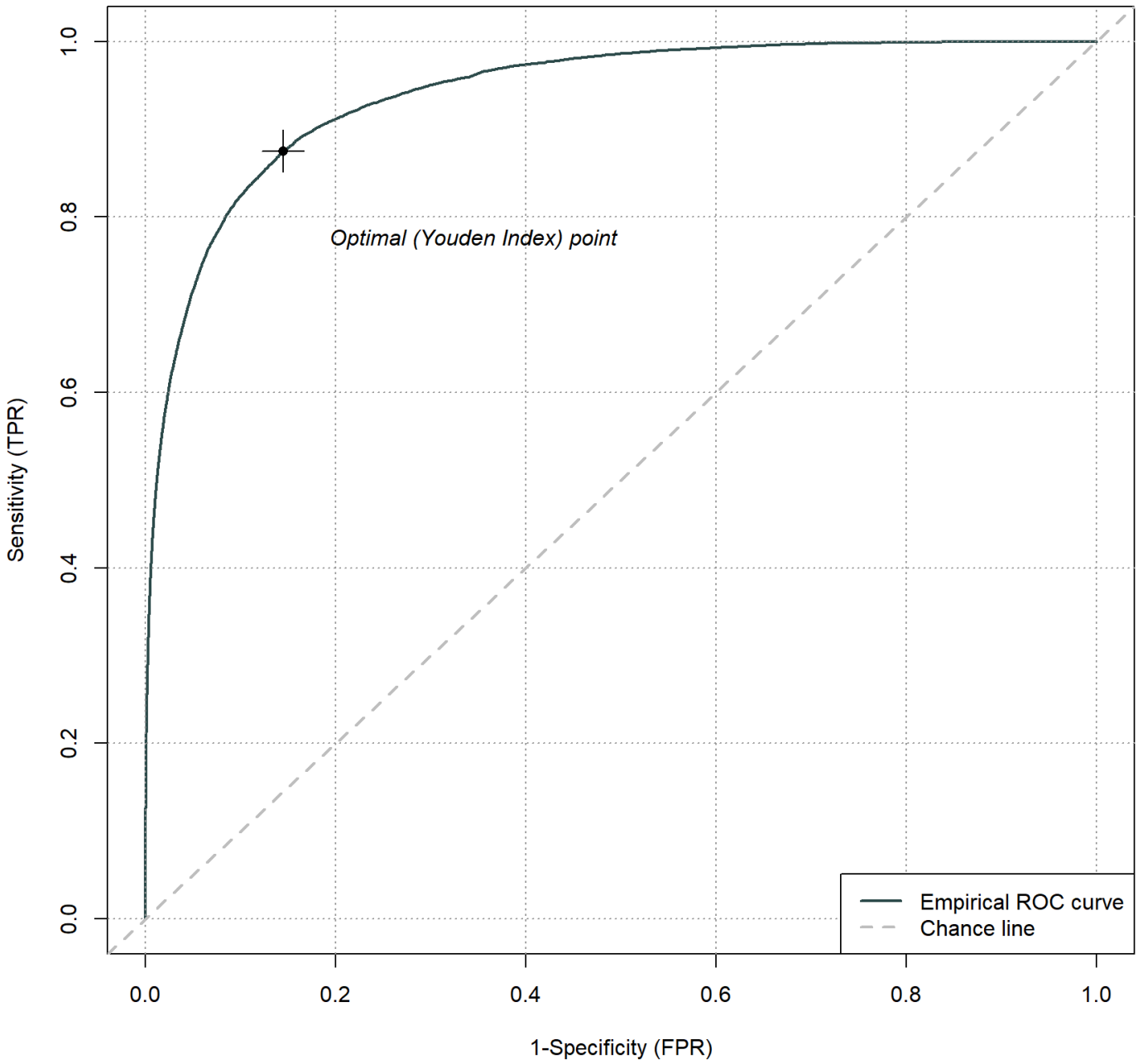
check convergence

```
hist.rug(apply(modelAsian$yhat.test |> pnorm(),2,mean),main="Distribution of prediction in test d
ata")
```

```
##    vars     n mean   sd median trimmed  mad min max range skew kurtosis se
## X1    1 30515 0.07 0.14   0.02    0.04 0.03   0   1     1 3.47    14.05  0
```




```
YoudenCutoff <- plot(rocit(score=apply(modelAsian$yhat.train |> pnorm(),2,mean) ,class=BTGINTT4$A
sian,negref =0))
```




```
macro_predictions <- split( modelAsian$yhat.train |> pnorm() %>% apply(c(1), mean) %>% t ,
                            rep(1:modelAsian$call$nchain , each=modelAsian$call$ndpost/modelAsian
$call$keepevery) )

mcmc_chains <- lapply(1:length(macro_predictions), function(i) {
  mcmc(macro_predictions[[i]])
}) %>%  as.mcmc.list

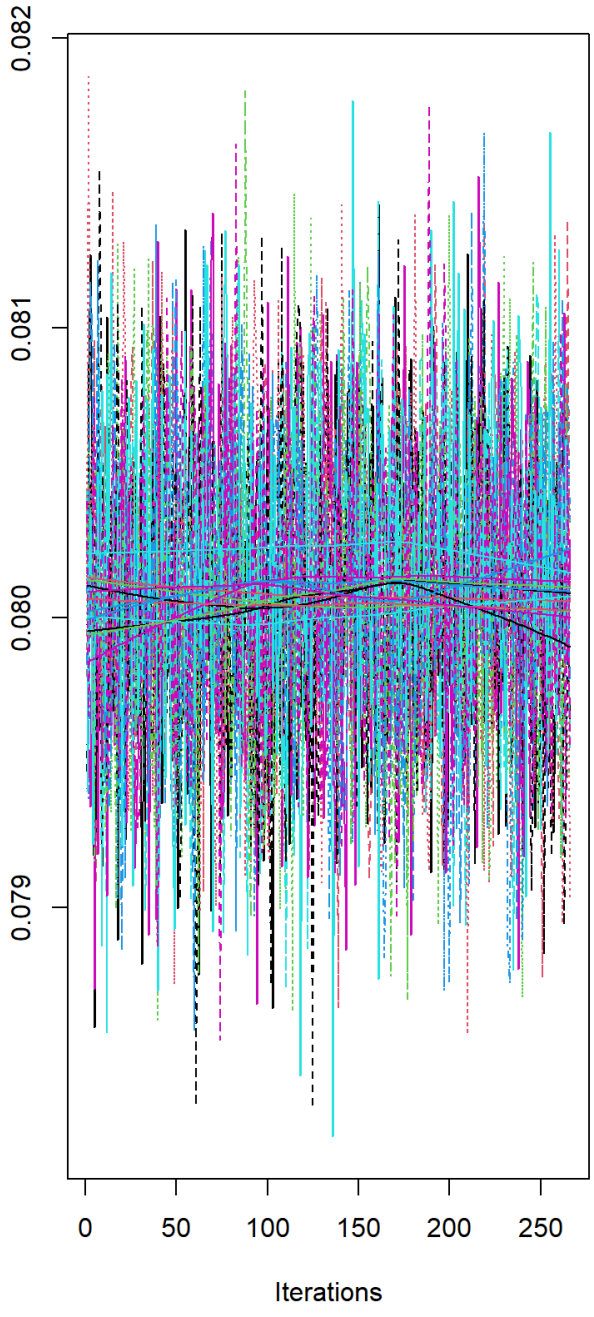
gelman.diag(mcmc_chains)
```

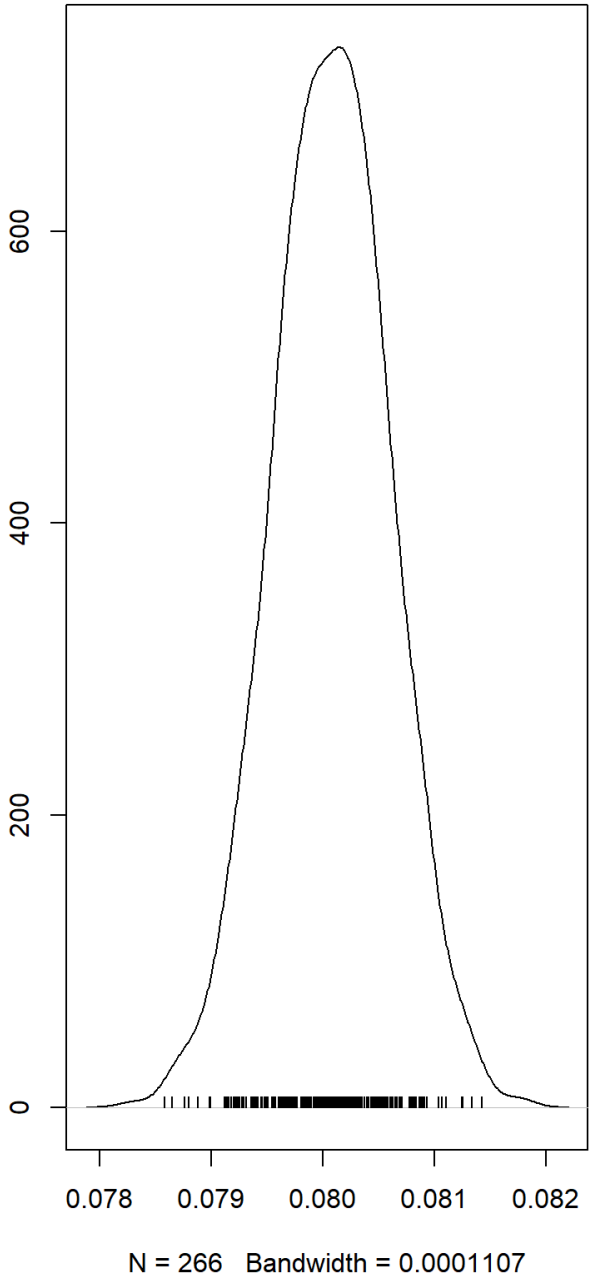
```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## [1,]          1       1.01
```

```
print(coda::effectiveSize(mcmc_chains))
```

```
##     var1
## 3251.875
```

```
plot(mcmc_chains)
```

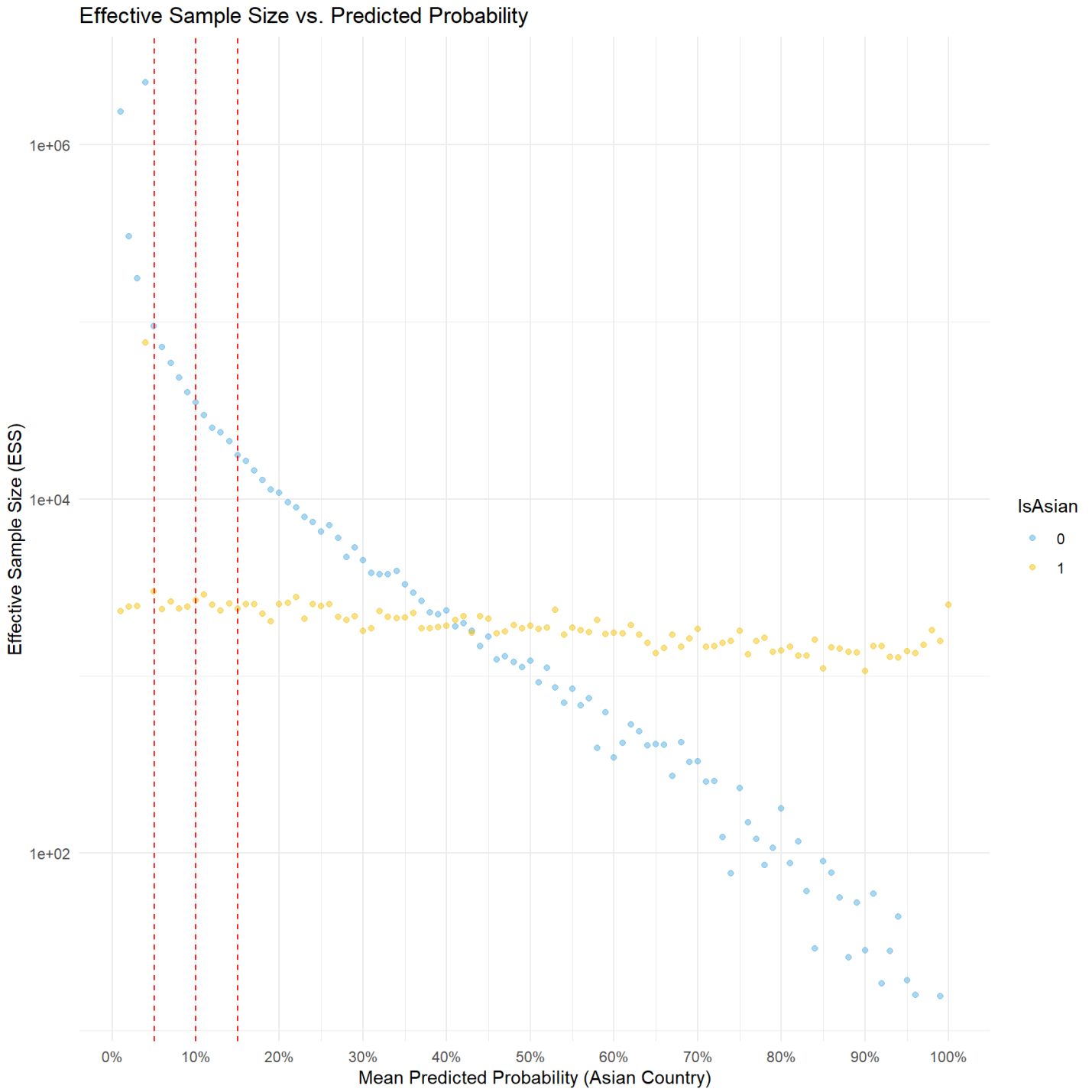




```
pred_prob_matrix <- modelAsian$yhat.train |> pnorm()
# Calculate ESS for each column (teacher) by looking across samples for each case
ess_values <- apply(pred_prob_matrix, 2, effectiveSize)
# Calculate the mean probability for each teacher
mean_probs <- colMeans(pred_prob_matrix)
# Create a data frame combining Teacher ID, Mean Probability, and ESS
teacher_diagnostics <- data.frame(
  IsAsian=modelAsian$fit$data@y,
  teacher_id = 1:length(mean_probs),
  mean_prob = mean_probs,
  ess = ess_values  )

cutoff_zone <- teacher_diagnostics[teacher_diagnostics$mean_prob >= YoudenCutoff$optimal[4]-0.05
&
                                   teacher_diagnostics$mean_prob <= YoudenCutoff$optimal[4]+0.05,
]

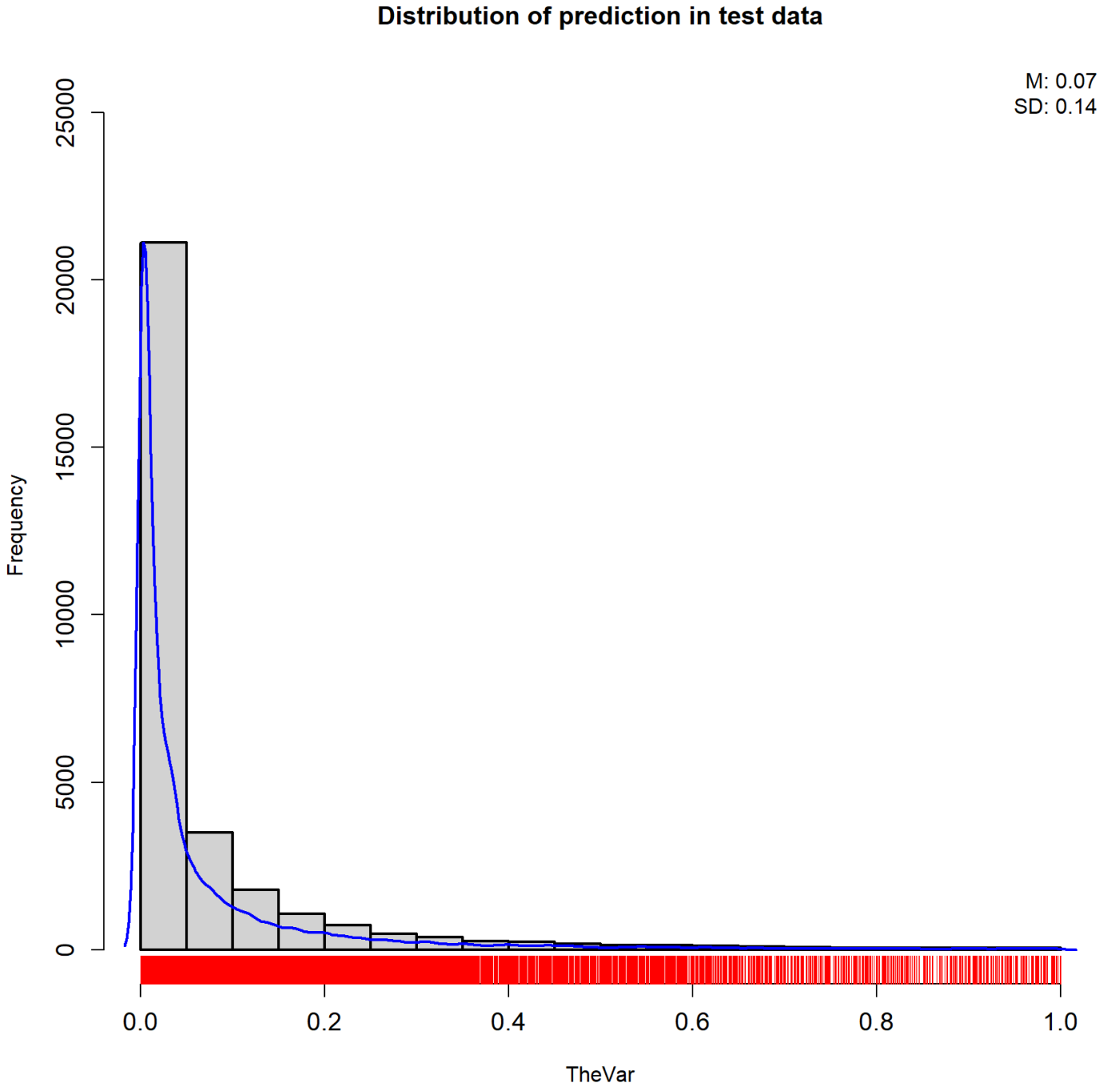
teacher_diagnostics |>
  mutate(mean_prob=cut(mean_prob,seq(0,1,0.01),include.lowest=T) |> factor() |> as.integer() ,
         mean_prob=mean_prob/100,
         IsAsian=factor(IsAsian) ) |>
  group_by(mean_prob,IsAsian) |>
  summarize(ess=sum(ess,na.rm=T) , .groups="drop") |>
  ggplot(aes(x = mean_prob, y = ess,color=IsAsian)) +
  geom_point(alpha = 0.5) +
  geom_vline(xintercept = c(max(YoudenCutoff$optimal[4]-0.05,0),
                            YoudenCutoff$optimal[4],
                            min(YoudenCutoff$optimal[4]+0.05,1)),
             linetype = "dashed", color = "red") +
  labs(title = "Effective Sample Size vs. Predicted Probability",
       x = "Mean Predicted Probability (Asian Country)",
       y = "Effective Sample Size (ESS)") +
  theme_minimal()+
  scale_x_continuous(breaks=seq(0,1,0.1),label=scales::percent)+
  scale_y_log10()
```



Results

```
hist.rug(apply(modelAsian$yhat.test |> pnorm(),2,mean),main="Distribution of prediction in test d
ata")
```

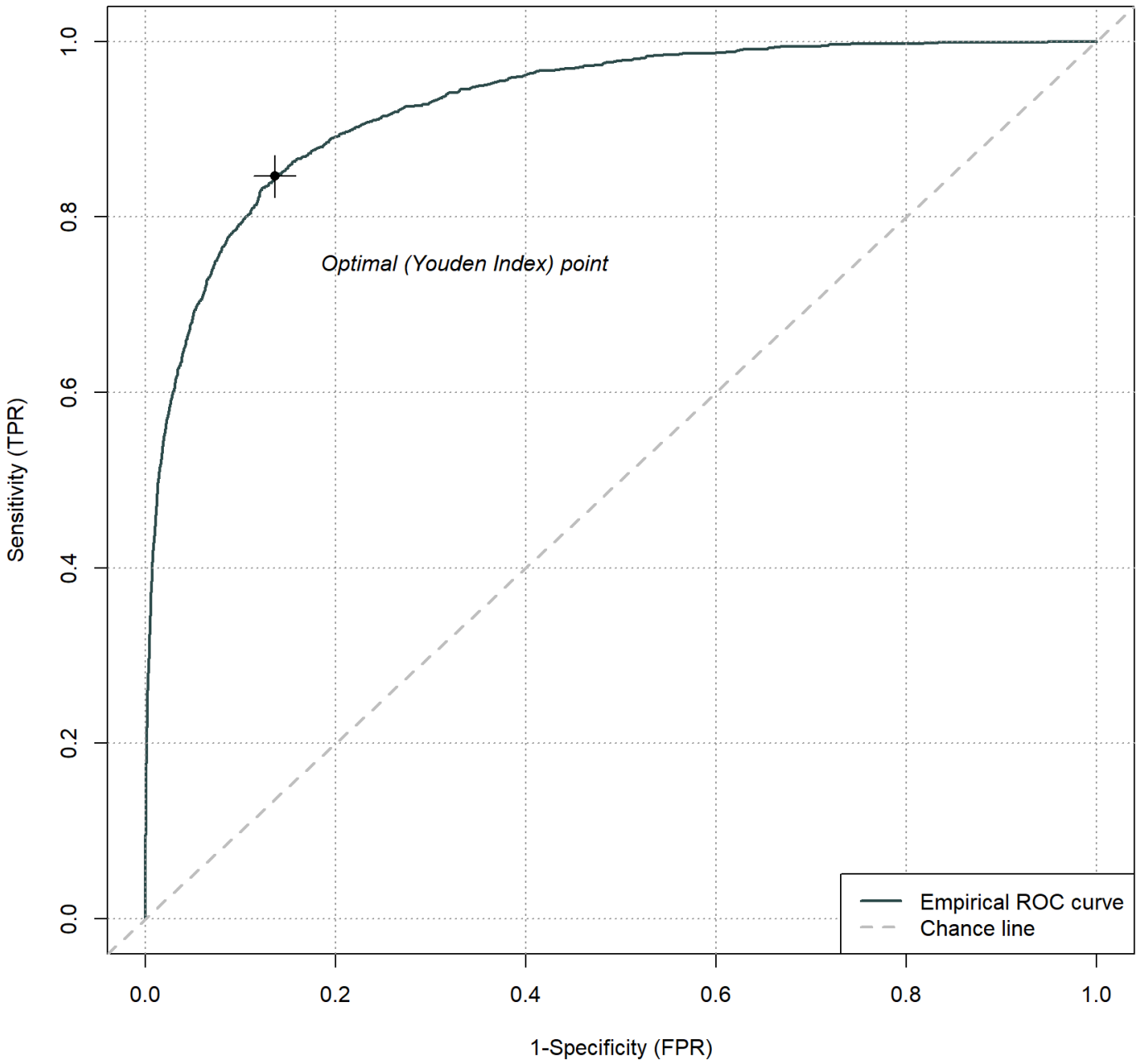
```
##    vars     n mean   sd median trimmed  mad min max range skew kurtosis se
## X1    1 30515 0.07 0.14   0.02    0.04 0.03   0   1     1 3.47    14.05  0
```




```
str( YoudenCutoff )
```

```
## List of 6
##  $ method                    : chr "empirical"
##  $ AUC                       : num 0.941
##  $ Cutoff                    : num [1:156146] Inf 1 1 1 1 ...
##  $ TPR/Sensitivity/Recall    : num [1:156146] 0 0.000079 0.000158 0.000237 0.000316 ...
##  $ FPR/1-Specificity         : num [1:156146] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.73 0.145 0.875 0.1
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

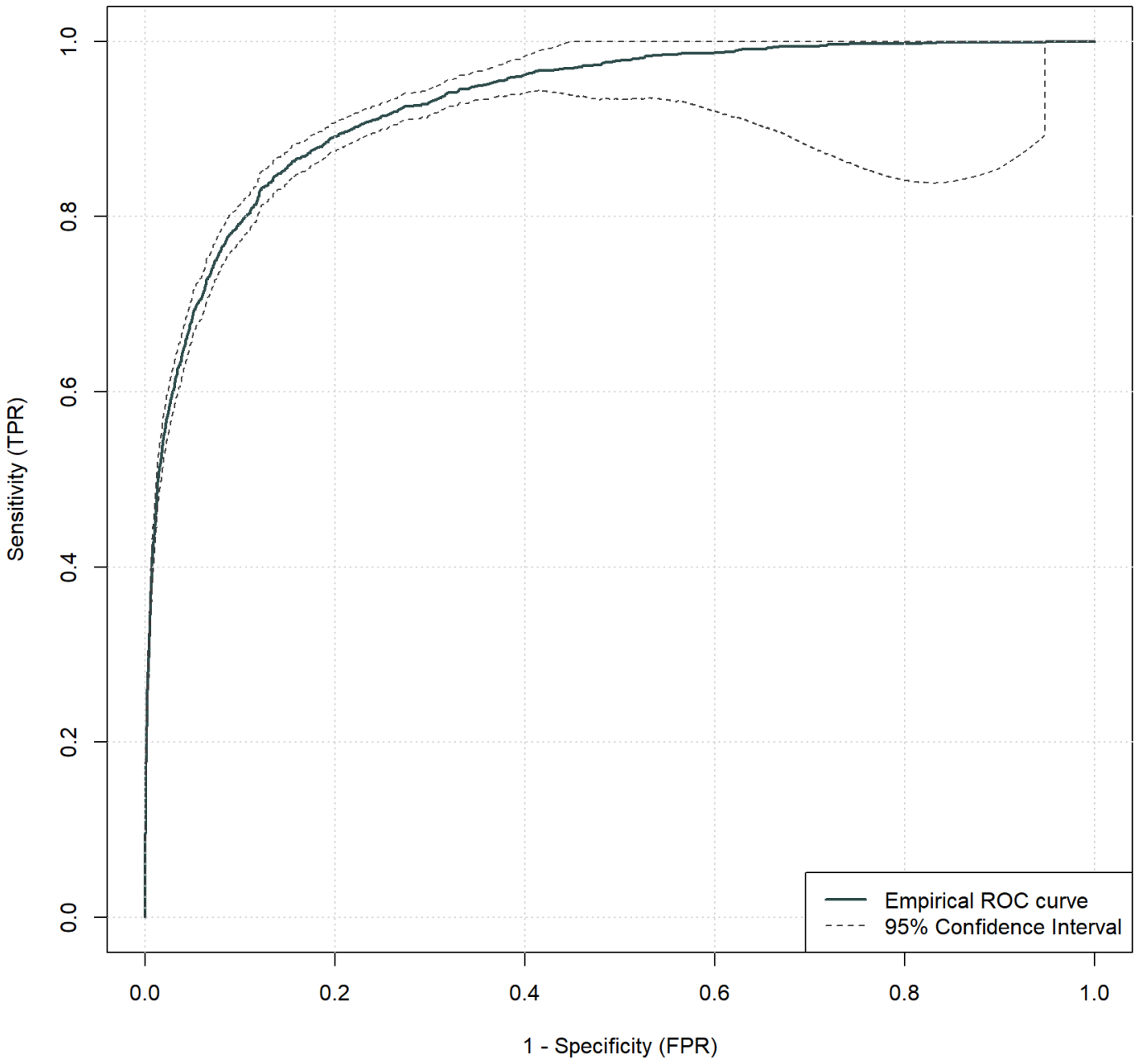
```
ROCit_obj_Asian <- rocit(score=apply(modelAsian$yhat.test |> pnorm(),2,mean) , class=Test$Asian,n
egref =0)
str( ROCIT_Asian <- plot(ROCit_obj_Asian) )
```




```
## List of 6
##  $ method                   : chr "empirical"
##  $ AUC                      : num 0.93
##  $ Cutoff                   : num [1:30079] Inf 1 0.999 0.998 0.998 ...
##  $ TPR/Sensitivity/Recall   : num [1:30079] 0 0.000698 0.001396 0.002094 0.002791 ...
##  $ FPR/1-Specificity        : num [1:30079] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.711 0.136 0.846 0.117
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

```
ciROC(ROCit_obj_Asian, method = "binomial") %>% plot(legend = TRUE)
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm): collapsing to unique
'x' values
```




```
ciAUC(ROCit_obj_Asian, delong = TRUE)
```

```
##
##    estimated AUC : 0.929870890371202
##    AUC estimation method : empirical
##
##    CI of AUC, delong method of variance used
##    confidence level = 95%
##    lower = 0.923158643915782     upper = 0.936583136826622
```

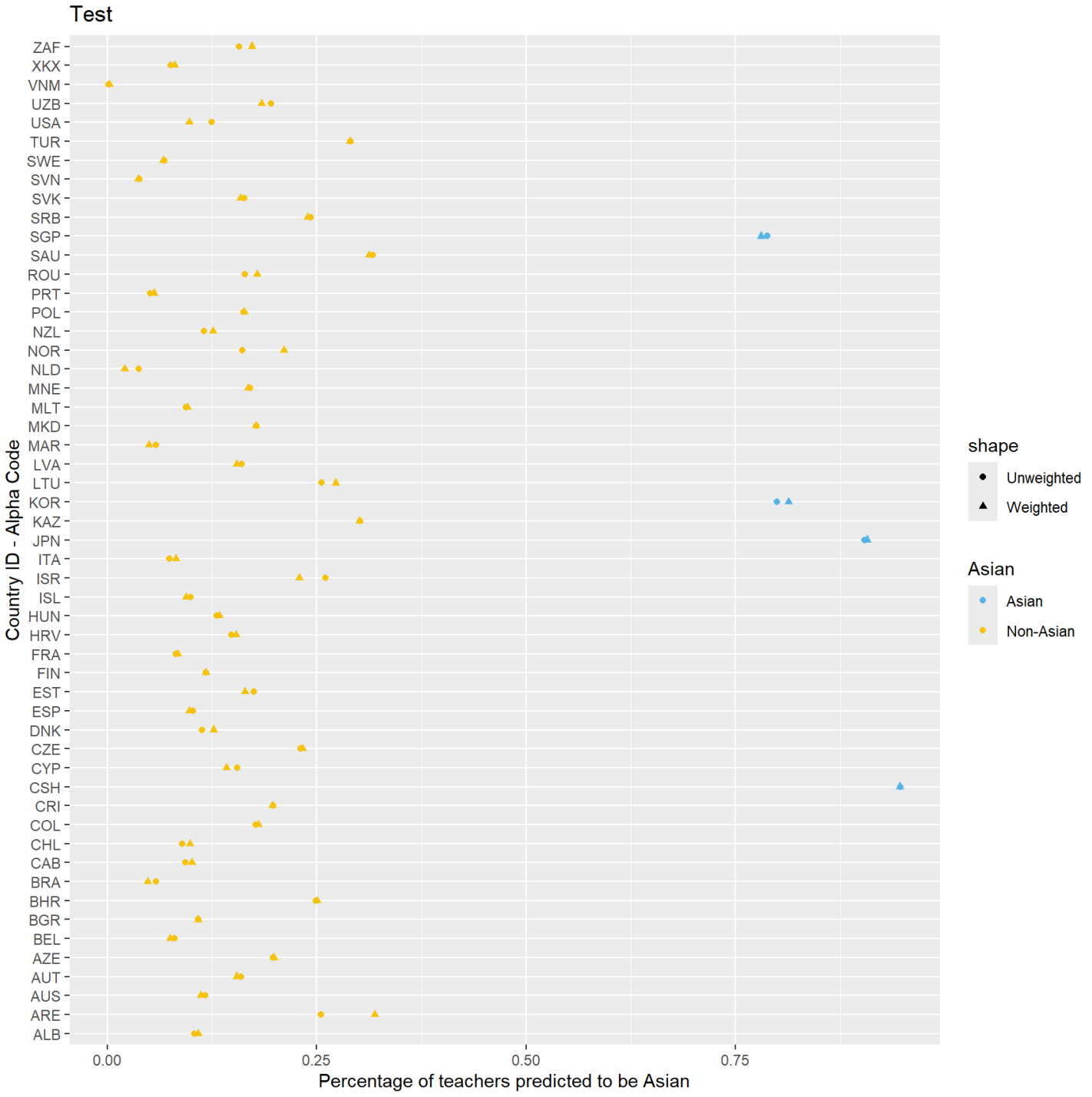
```
save(ROCIT_Asian,ROCit_obj_Asian,YoudenCutoff,file="ROCIT_Asian_Test.rData")
```

Tables for classification

```
Test %>%
  transmute(CNTRY,predicted=apply(modelAsian$yhat.test |> pnorm(),2,mean)>=YoudenCutoff$optimal
[4]) %>%
  table
```

```
##      predicted
## CNTRY FALSE TRUE
##   ALB   362   42
##   ARE   327  112
##   AUS   278   37
##   AUT   379   72
##   AZE   349   86
##   BEL  1089   95
##   BGR   343   42
##   BHR   235   78
##   BRA   292   18
##   CAB   175   18
##   CHL   214   21
##   COL   260   56
##   CRI   247   61
##   CSH    22  391
##   CYP   158   29
##   CZE  1829  549
##   DNK   227   29
##   ESP  4581  524
##   EST   282   60
##   FIN   396   53
##   FRA   268   24
##   HRV   269   47
##   HUN   352   53
##   ISL   136   15
##   ISR   247   87
##   ITA   362   29
##   JPN    35  331
##   KAZ  2350 1017
##   KOR    63  251
##   LTU   322  111
##   LVA   344   66
##   MAR  1992  124
##   MKD   332   72
##   MLT   230   24
##   MNE   117   24
##   NLD   128    5
##   NOR   130   25
##   NZL   115   15
##   POL   318   62
##   PRT   350   19
##   ROU   330   65
##   SAU   228  106
##   SGP    72  268
##   SRB   314  101
##   SVK   357   70
##   SVN   273   11
##   SWE   285   21
##   TUR   677  277
##   USA   175   25
##   UZB   369   90
##   VNM   449    1
##   XKX   304   25
##   ZAF   289   54
```

```
Test %>%
  mutate(predicted = apply(modelAsian$yhat.test |> pnorm(),2,mean),
         Asian=factor(Asian,1:0,c("Asian","Non-Asian")) ) %>%
  group_by(CNTRY,Asian) %>%
  summarize(Unwg_pred=mean(predicted>=YoudenCutoff$optimal[4]),
            Wgt_pred =weighted.mean(predicted>=YoudenCutoff$optimal[4],unlabelled(TCHWGT)) ,
            .groups="drop") %>%
  ggplot(aes(y=CNTRY,color=Asian))+
    geom_point(aes(x=Unwg_pred,shape="Unweighted"))+
    geom_point(aes(x=Wgt_pred,shape="Weighted"))+
    labs(x="Percentage of teachers predicted to be Asian",title="Test")
```



remove variables to create space in RAM

```
rm(modelAsian, ROCit_obj_Asian,ROCIT_Asian,mcmc_chains,macro_predictions)
gc()
```

```
##             used   (Mb) gc trigger    (Mb)   max used    (Mb)
## Ncells   4041228  215.9    7386087   394.5    7386087   394.5
## Vcells 526184339 4014.5 2914403818 22235.2 3427414240 26149.1
```

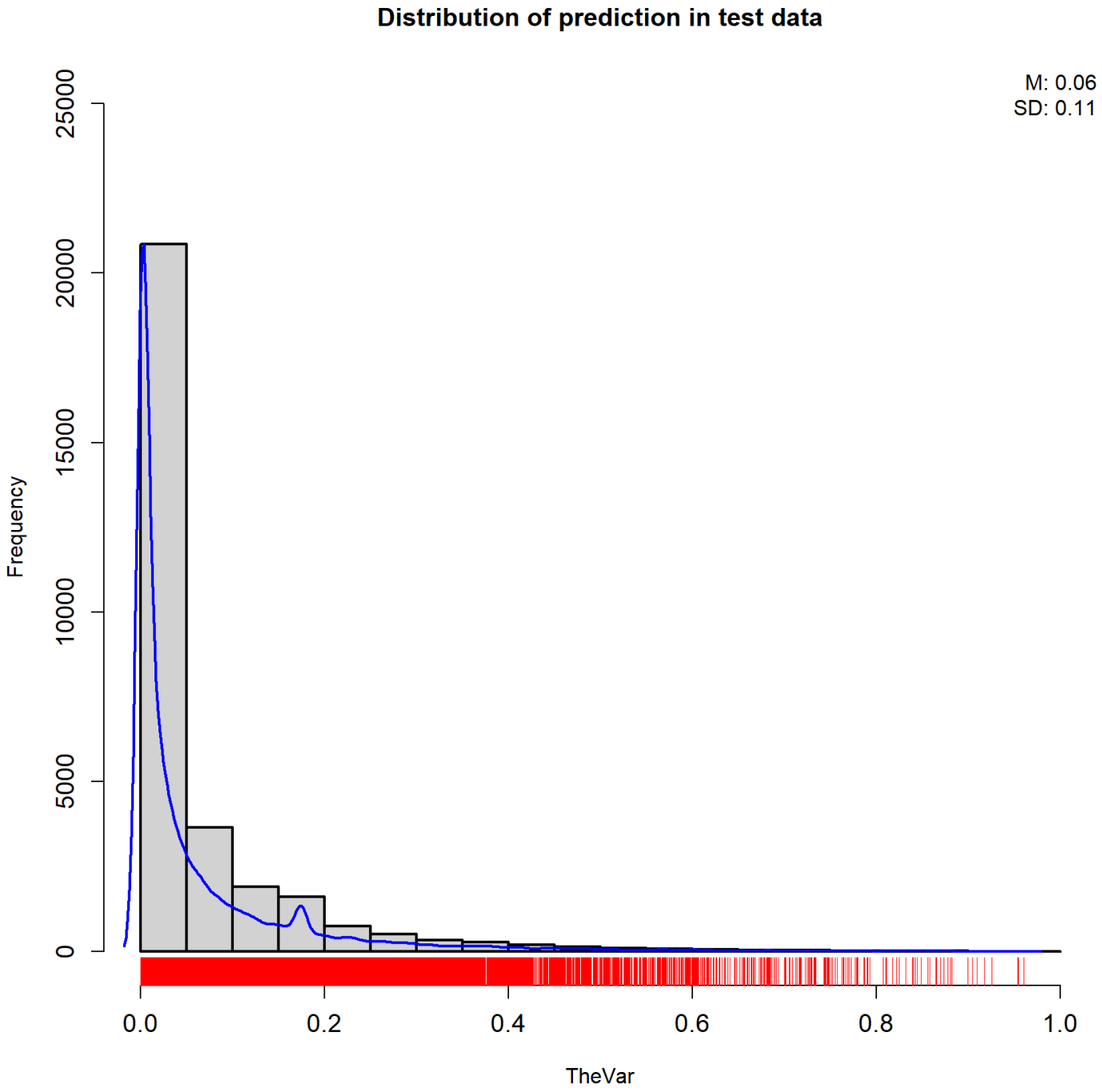
Models for Nordic Countries without Island

```
load(file="modelIsl.Rdata")
```

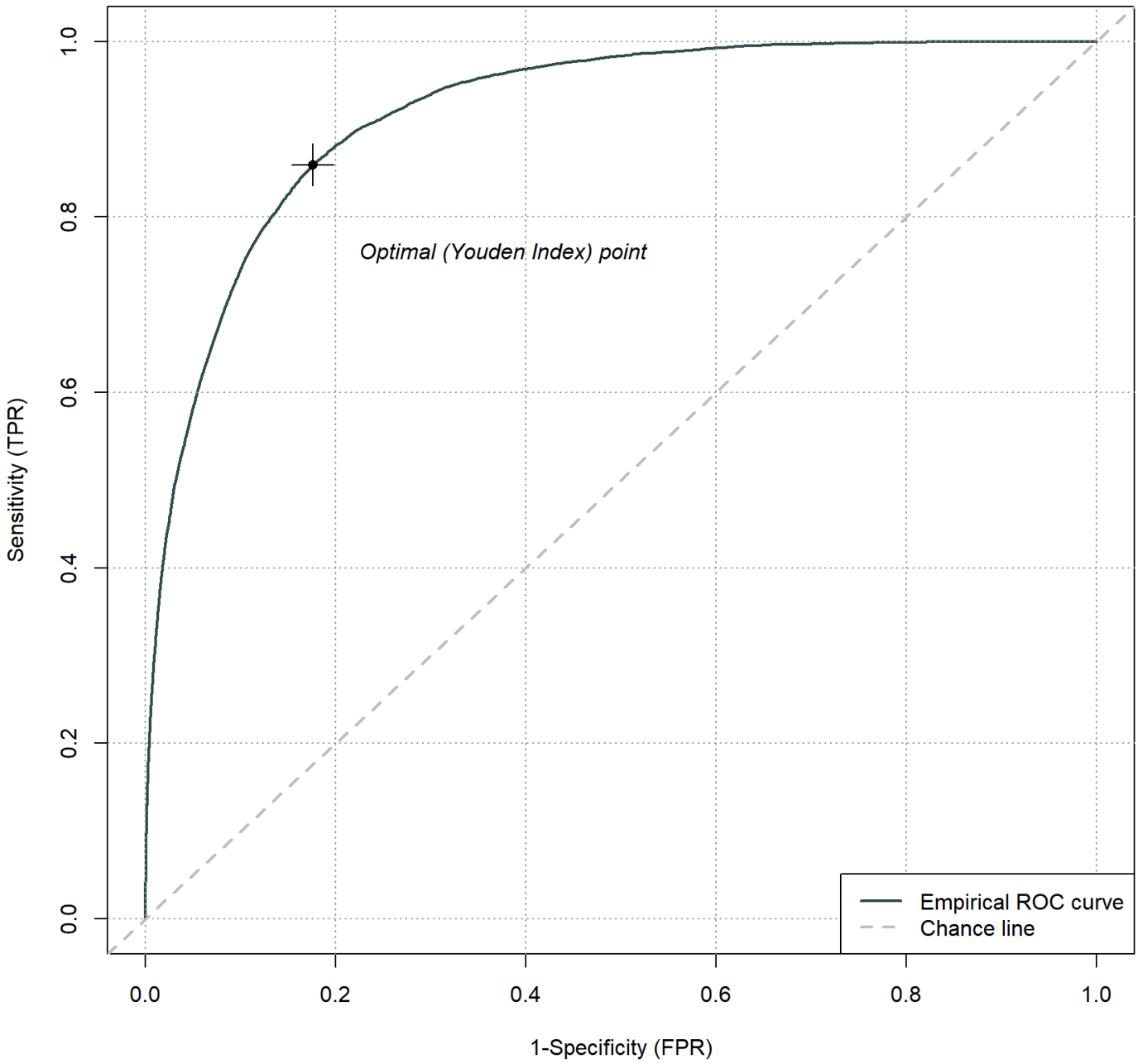
check convergence

```
hist.rug(apply(modelIsl$yhat.test |> pnorm(),2,mean),main="Distribution of prediction in test dat
a")
```

```
##    vars     n mean   sd median trimmed  mad min  max range skew kurtosis se
## X1    1 30515 0.06 0.11   0.02    0.04 0.02   0 0.96  0.96  3.1    12.42  0
```




```
YoudenCutoff <- plot(rocit(score=apply(modelIsl$yhat.train |> pnorm(),2,mean) ,class=BTGINTT4$Nor
dic,negref =0))
```




```
macro_predictions <- split( modelIsl$yhat.train |> pnorm() %>% apply(c(1), mean) %>% t ,
                           rep(1:modelIsl$call$nchain , each=modelIsl$call$ndpost/modelIsl$call
$keepevery) )

mcmc_chains <- lapply(1:length(macro_predictions), function(i) {
  mcmc(macro_predictions[[i]])
}) %>%  as.mcmc.list

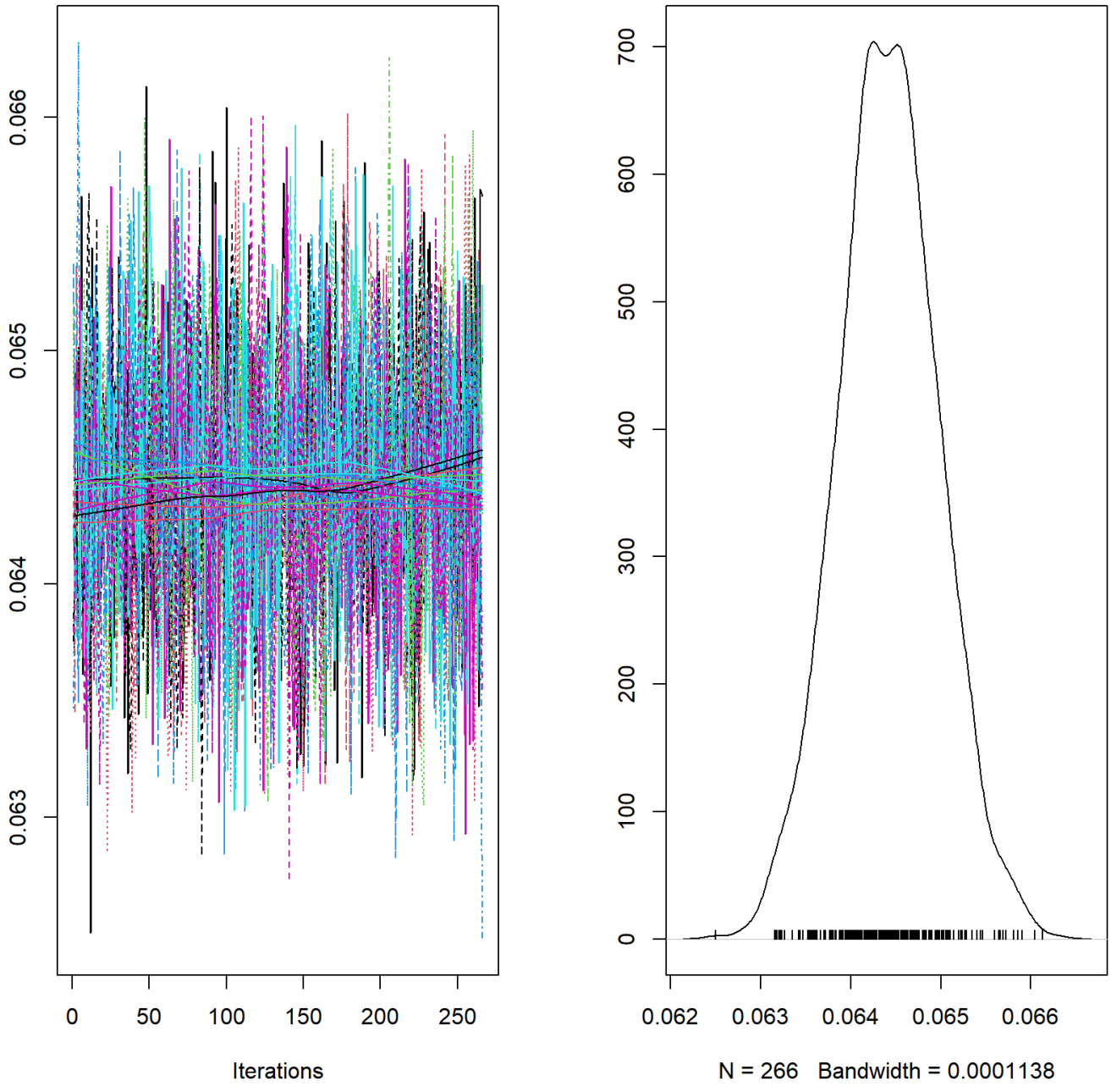
gelman.diag(mcmc_chains)
```

```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## [1,]          1       1.01
```

```
print(coda::effectiveSize(mcmc_chains))
```

```
##     var1
## 3191.026
```

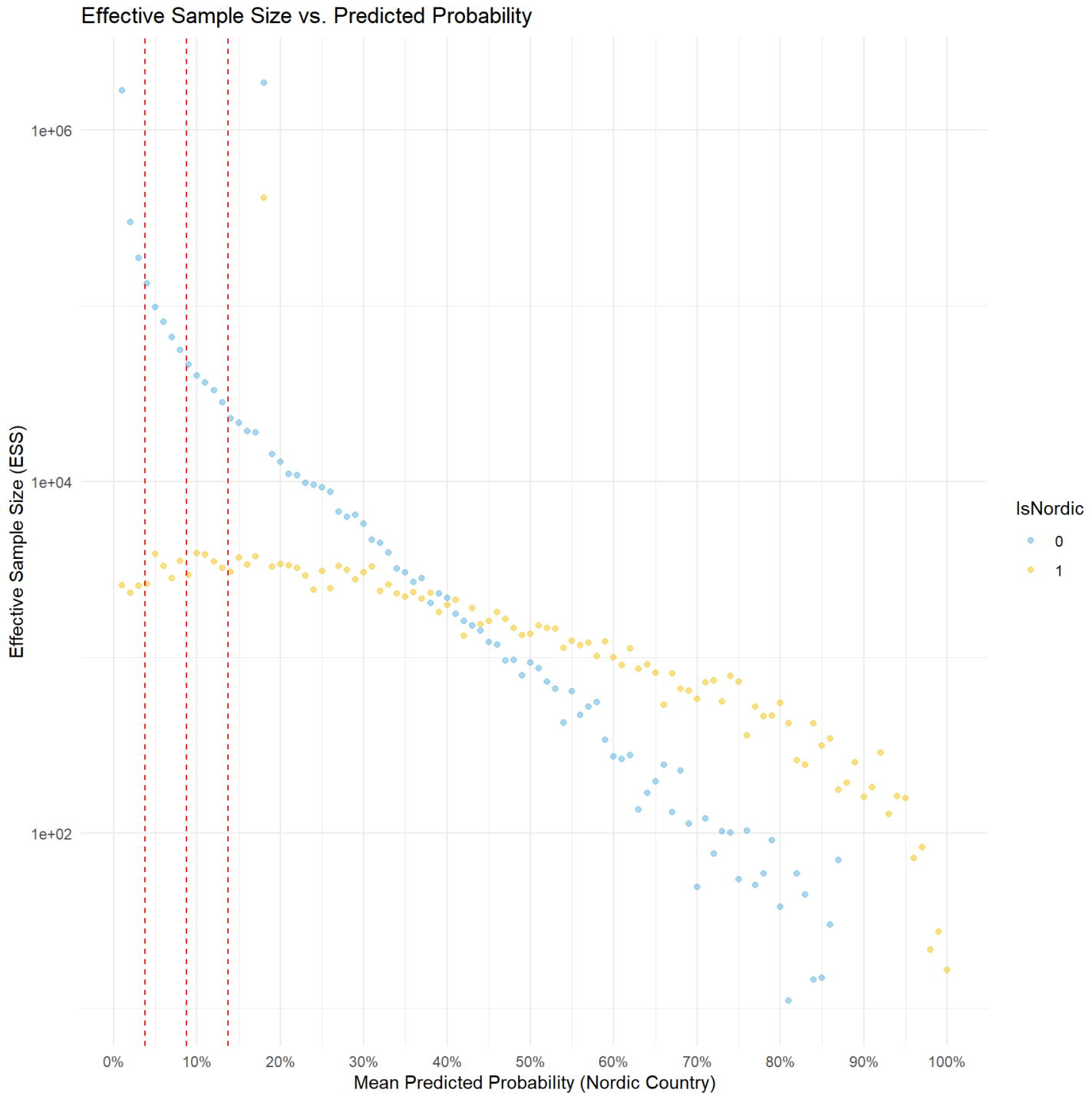
```
plot(mcmc_chains)
```



```
pred_prob_matrix <- modelIsl$yhat.train |> pnorm()
# Calculate ESS for each column (teacher) by looking across samples for each case
ess_values <- apply(pred_prob_matrix, 2, effectiveSize)
# Calculate the mean probability for each teacher
mean_probs <- colMeans(pred_prob_matrix)
# Create a data frame combining Teacher ID, Mean Probability, and ESS
teacher_diagnostics <- data.frame(
  IsNordic=modelIsl$fit$data@y,
  teacher_id = 1:length(mean_probs),
  mean_prob = mean_probs,
  ess = ess_values  )

cutoff_zone <- teacher_diagnostics[teacher_diagnostics$mean_prob >= YoudenCutoff$optimal[4]-0.05
&
                                   teacher_diagnostics$mean_prob <= YoudenCutoff$optimal[4]+0.05,
]

teacher_diagnostics |>
  mutate(mean_prob=cut(mean_prob,seq(0,1,0.01),include.lowest=T) |> factor() |> as.integer() ,
         mean_prob=mean_prob/100,
         IsNordic=factor(IsNordic) ) |>
  group_by(mean_prob,IsNordic) |>
  summarize(ess=sum(ess,na.rm=T) , .groups="drop") |>
  ggplot(aes(x = mean_prob, y = ess,color=IsNordic)) +
  geom_point(alpha = 0.5) +
  geom_vline(xintercept = c(max(YoudenCutoff$optimal[4]-0.05,0),
                            YoudenCutoff$optimal[4],
                            min(YoudenCutoff$optimal[4]+0.05,1)),
             linetype = "dashed", color = "red") +
  labs(title = "Effective Sample Size vs. Predicted Probability",
       x = "Mean Predicted Probability (Nordic Country)",
       y = "Effective Sample Size (ESS)") +
  theme_minimal()+
  scale_x_continuous(breaks=seq(0,1,0.1),label=scales::percent)+
  scale_y_log10()
```

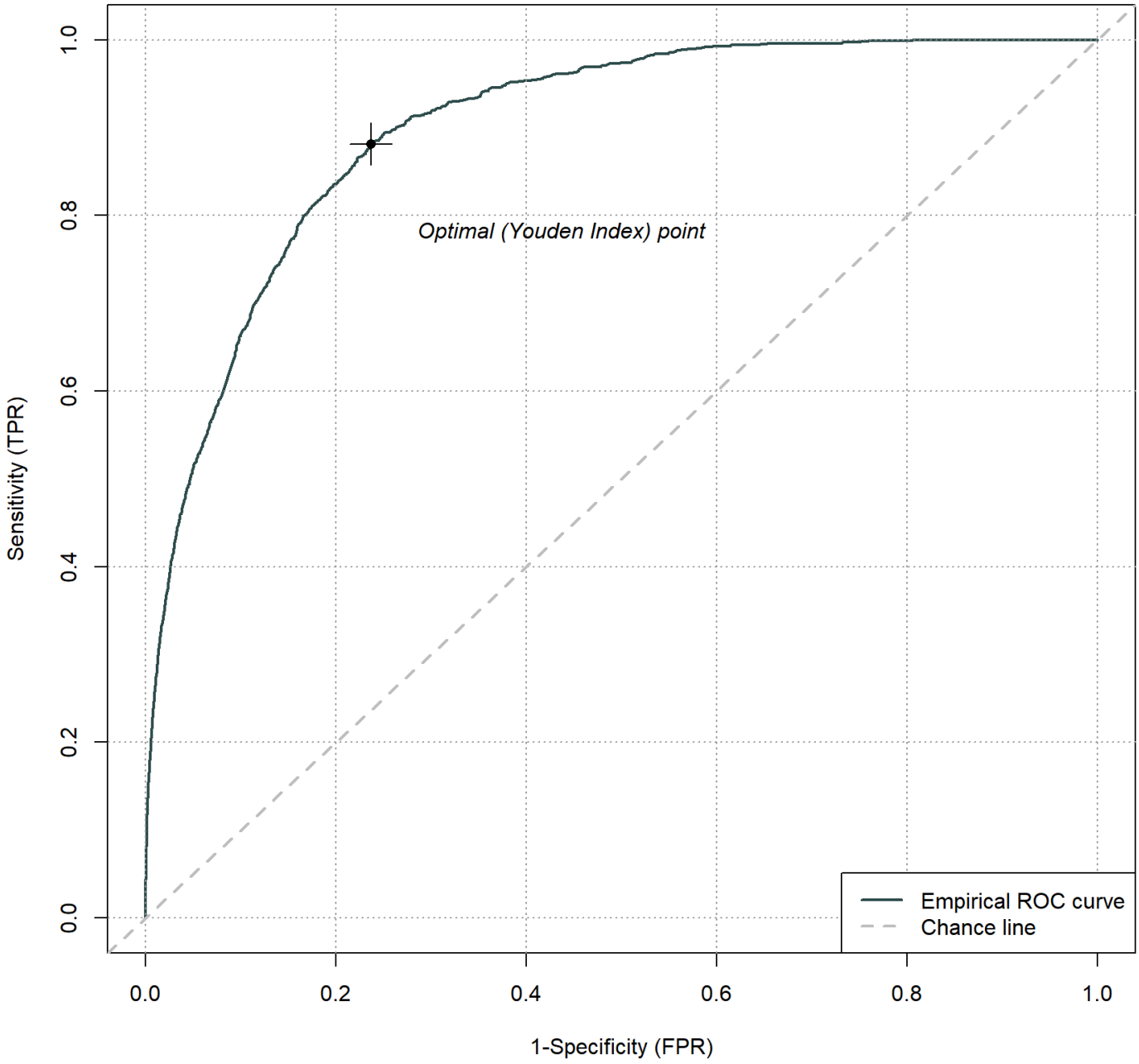


Results Nordic

```
str( YoudenCutoff )
```

```
## List of 6
##  $ method                    : chr "empirical"
##  $ AUC                       : num 0.919
##  $ Cutoff                    : num [1:156145] Inf 0.993 0.986 0.973 0.968 ...
##  $ TPR/Sensitivity/Recall    : num [1:156145] 0.00 9.79e-05 1.96e-04 2.94e-04 3.92e-04 ...
##  $ FPR/1-Specificity         : num [1:156145] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.6835 0.176 0.8594 0.0875
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

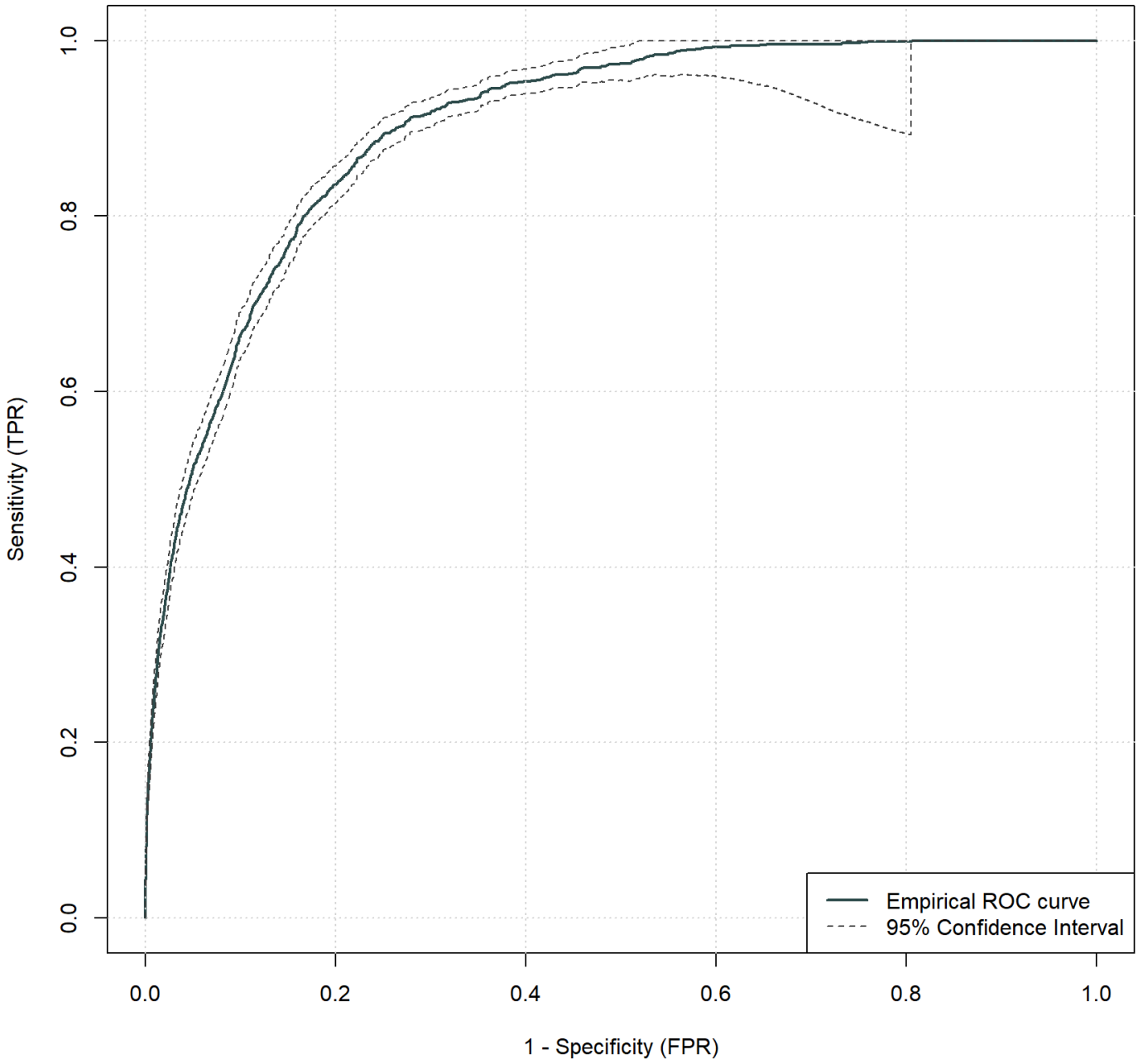
```
ROCit_obj_NordicIsl    <- rocit(score=apply(modelIsl$yhat.test |> pnorm(),2,mean) ,class=Test$Nor
dic,negref =0)
str( ROCIT_NordicIsl <- plot(ROCit_obj_NordicIsl) )
```




```
## List of 6
##  $ method                    : chr "empirical"
##  $ AUC                       : num 0.9
##  $ Cutoff                    : num [1:30078] Inf 0.961 0.954 0.954 0.926 ...
##  $ TPR/Sensitivity/Recall    : num [1:30078] 0 0.000858 0.001715 0.002573 0.003431 ...
##  $ FPR/1-Specificity         : num [1:30078] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.6449 0.2367 0.8816 0.0681
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

```
ciROC(ROCit_obj_NordicIsl, method = "binomial") %>% plot(legend = TRUE)
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm): collapsing to unique
'x' values
```




```
ciAUC(ROCit_obj_NordicIsl, delong = TRUE)
```

```
##
##    estimated AUC : 0.900185439123316
##    AUC estimation method : empirical
##
##    CI of AUC, delong method of variance used
##    confidence level = 95%
##    lower = 0.892451086041967     upper = 0.907919792204664
```

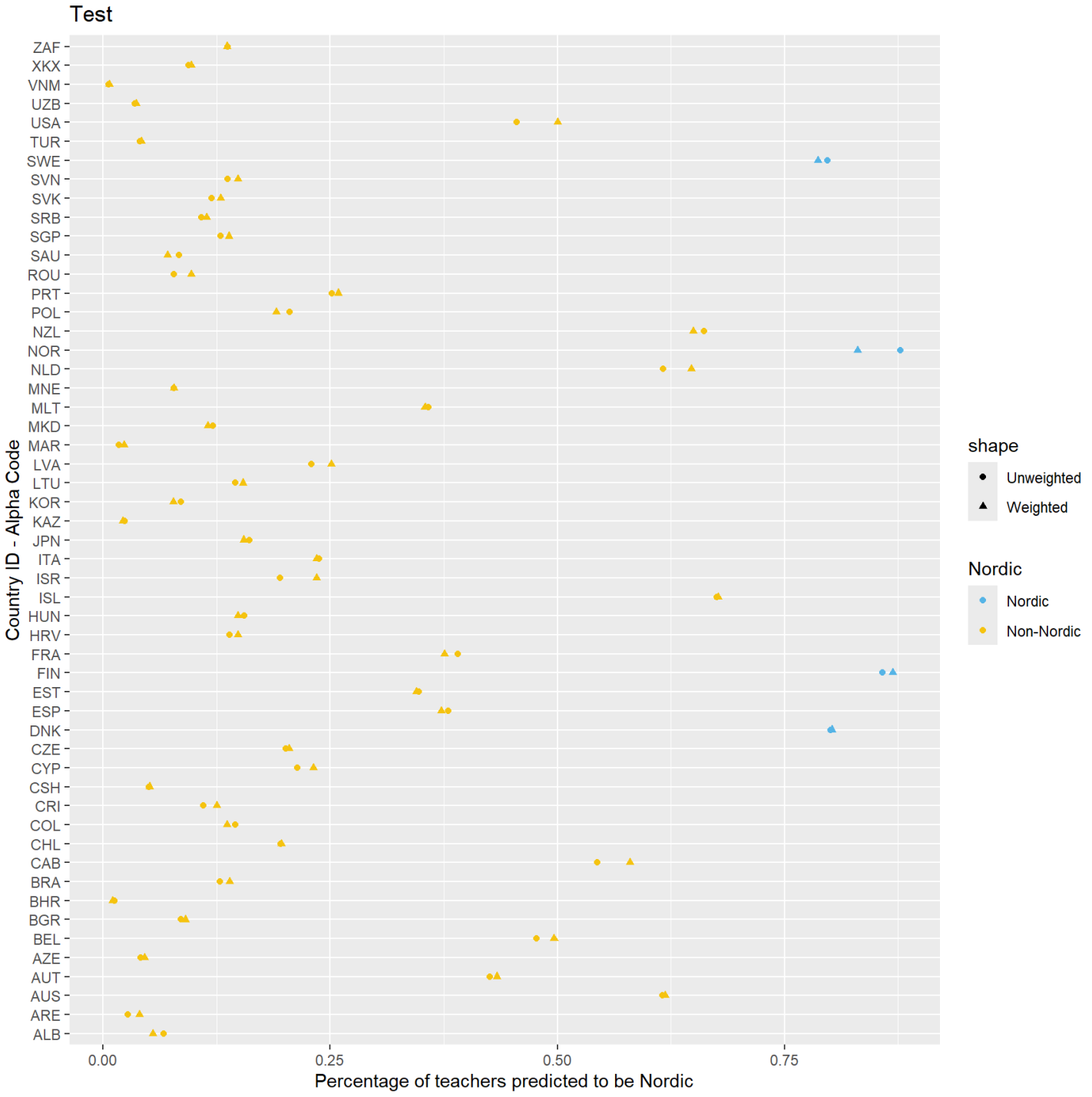
```
save(ROCIT_NordicIsl,ROCit_obj_NordicIsl,YoudenCutoff,file="ROCIT_NordicIsl_Test.rData")
```

Tables for classification

```
Test %>%
  transmute(CNTRY,predicted=apply(modelIsl$yhat.test |> pnorm(),2,mean)>=YoudenCutoff$optimal[4])
%>%
  table
```

```
##      predicted
## CNTRY FALSE TRUE
##   ALB   377   27
##   ARE   427   12
##   AUS   121  194
##   AUT   259  192
##   AZE   417   18
##   BEL   619  565
##   BGR   352   33
##   BHR   309    4
##   BRA   270   40
##   CAB    88  105
##   CHL   189   46
##   COL   270   46
##   CRI   274   34
##   CSH   392   21
##   CYP   147   40
##   CZE  1900  478
##   DNK    51  205
##   ESP  3164 1941
##   EST   223  119
##   FIN    64  385
##   FRA   178  114
##   HRV   272   44
##   HUN   342   63
##   ISL    49  102
##   ISR   269   65
##   ITA   298   93
##   JPN   307   59
##   KAZ  3287   80
##   KOR   287   27
##   LTU   370   63
##   LVA   316   94
##   MAR  2078   38
##   MKD   355   49
##   MLT   163   91
##   MNE   130   11
##   NLD    51   82
##   NOR    19  136
##   NZL    44   86
##   POL   302   78
##   PRT   276   93
##   ROU   364   31
##   SAU   306   28
##   SGP   296   44
##   SRB   370   45
##   SVK   376   51
##   SVN   245   39
##   SWE    62  244
##   TUR   915   39
##   USA   109   91
##   UZB   443   16
##   VNM   447    3
##   XKX   298   31
##   ZAF   296   47
```

```
Test %>%
  mutate(predicted = apply(modelIsl$yhat.test |> pnorm(),2,mean),
         Nordic=factor(Nordic,1:0,c("Nordic","Non-Nordic")) ) %>%
  group_by(CNTRY,Nordic) %>%
  summarize(Unwg_pred=         mean(predicted>=YoudenCutoff$optimal[4]),
            Wgt_pred =weighted.mean(predicted>=YoudenCutoff$optimal[4],unlabelled(TCHWGT)) ,
            .groups="drop") %>%
  ggplot(aes(y=CNTRY,color=Nordic))+
    geom_point(aes(x=Unwg_pred,shape="Unweighted"))+
    geom_point(aes(x=Wgt_pred,shape="Weighted"))+
    labs(x="Percentage of teachers predicted to be Nordic",title="Test")
```



remove variables to create space in RAM

```
rm(modelIsl,ROCIT_NordicIsl,ROCit_obj_NordicIsl,mcmc_chains,macro_predictions)
gc()
```

```
##             used   (Mb) gc trigger    (Mb)   max used    (Mb)
## Ncells   4041260  215.9    7386090   394.5    7386090   394.5
## Vcells 526216849 4014.8 2797891665 21346.3 3497362074 26682.8
```

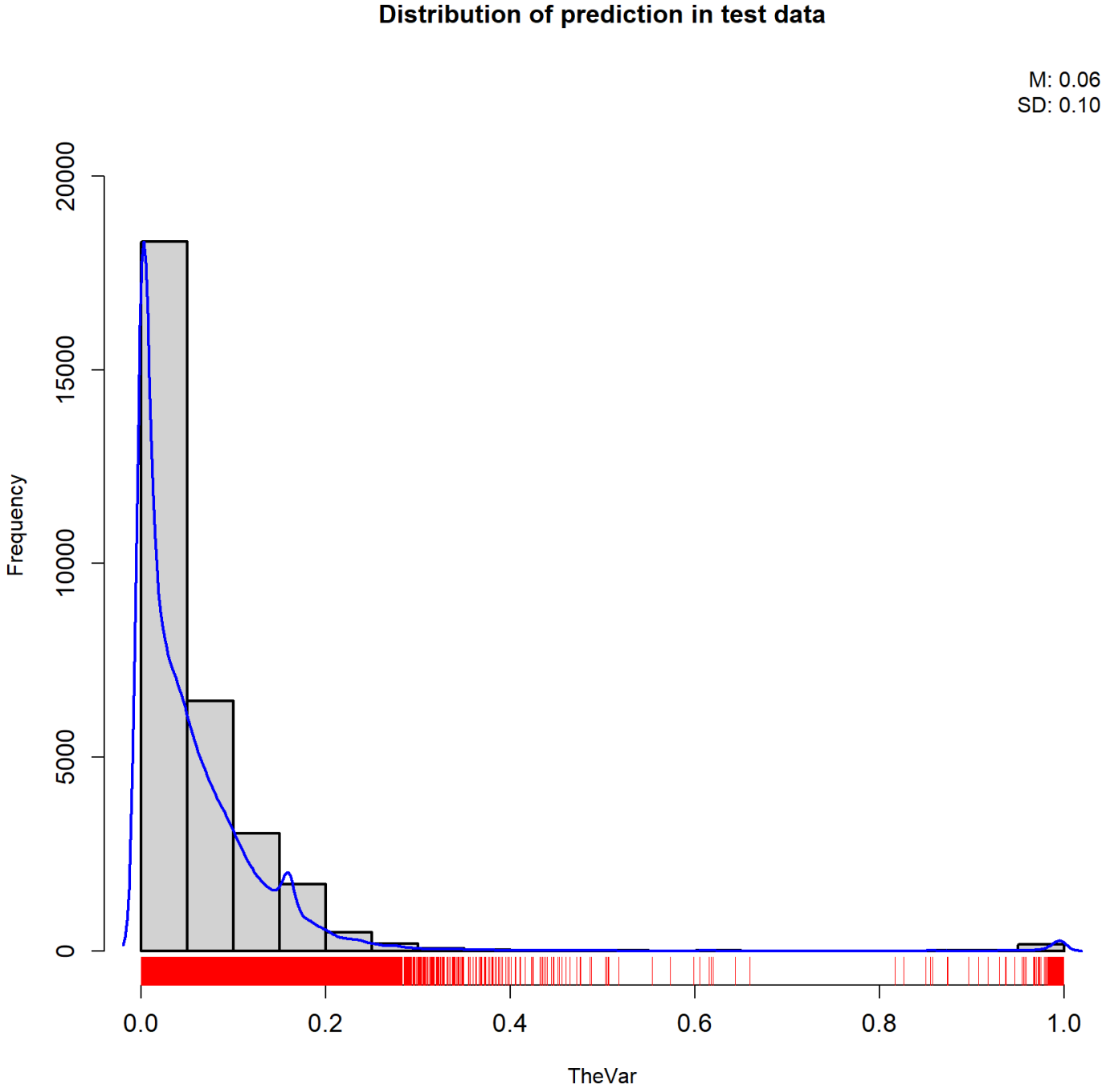
Original Random country Models

```
load(file="ModelRand2.Rdata")
```

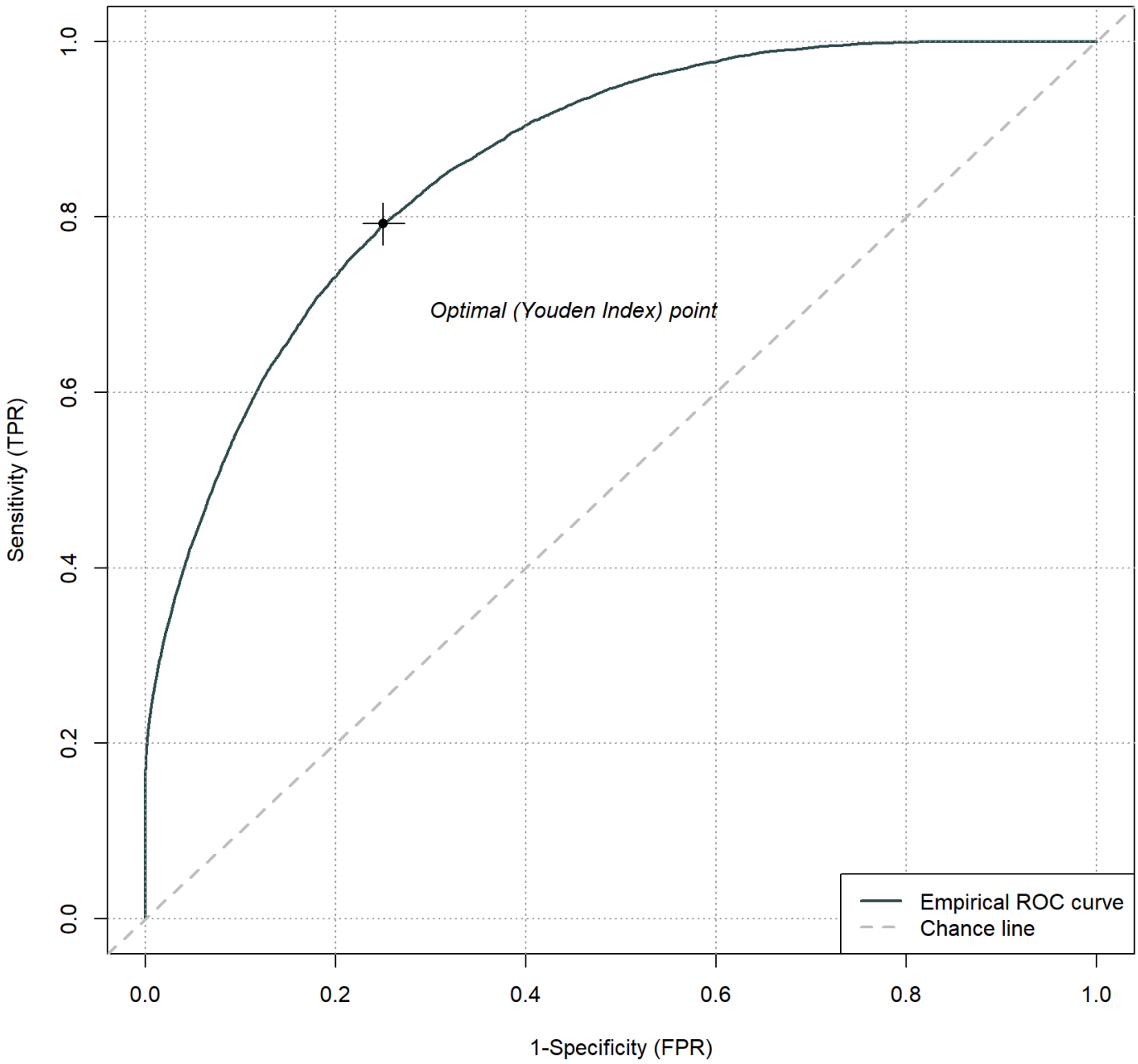
check convergence

```
hist.rug(apply(modelRand2$yhat.test |> pnorm(),2,mean),main="Distribution of prediction in test d
ata")
```

```
##    vars     n mean  sd median trimmed  mad min max range skew kurtosis se
## X1    1 30515 0.06 0.1   0.03    0.04 0.05   0   1     1 6.18    54.58  0
```




```
YoudenCutoff <- plot(rocit(score=apply(modelRand2$yhat.train |> pnorm(),2,mean) ,class=BTGINTT4$R
andom2,negref =0))
```




```
macro_predictions <- split( modelRand2$yhat.train |> pnorm() %>% apply(c(1), mean) %>% t ,
                           rep(1:modelRand2$call$nchain , each=modelRand2$call$ndpost/modelRand2
$call$keepevery) )

mcmc_chains <- lapply(1:length(macro_predictions), function(i) {
  mcmc(macro_predictions[[i]])
}) %>%  as.mcmc.list

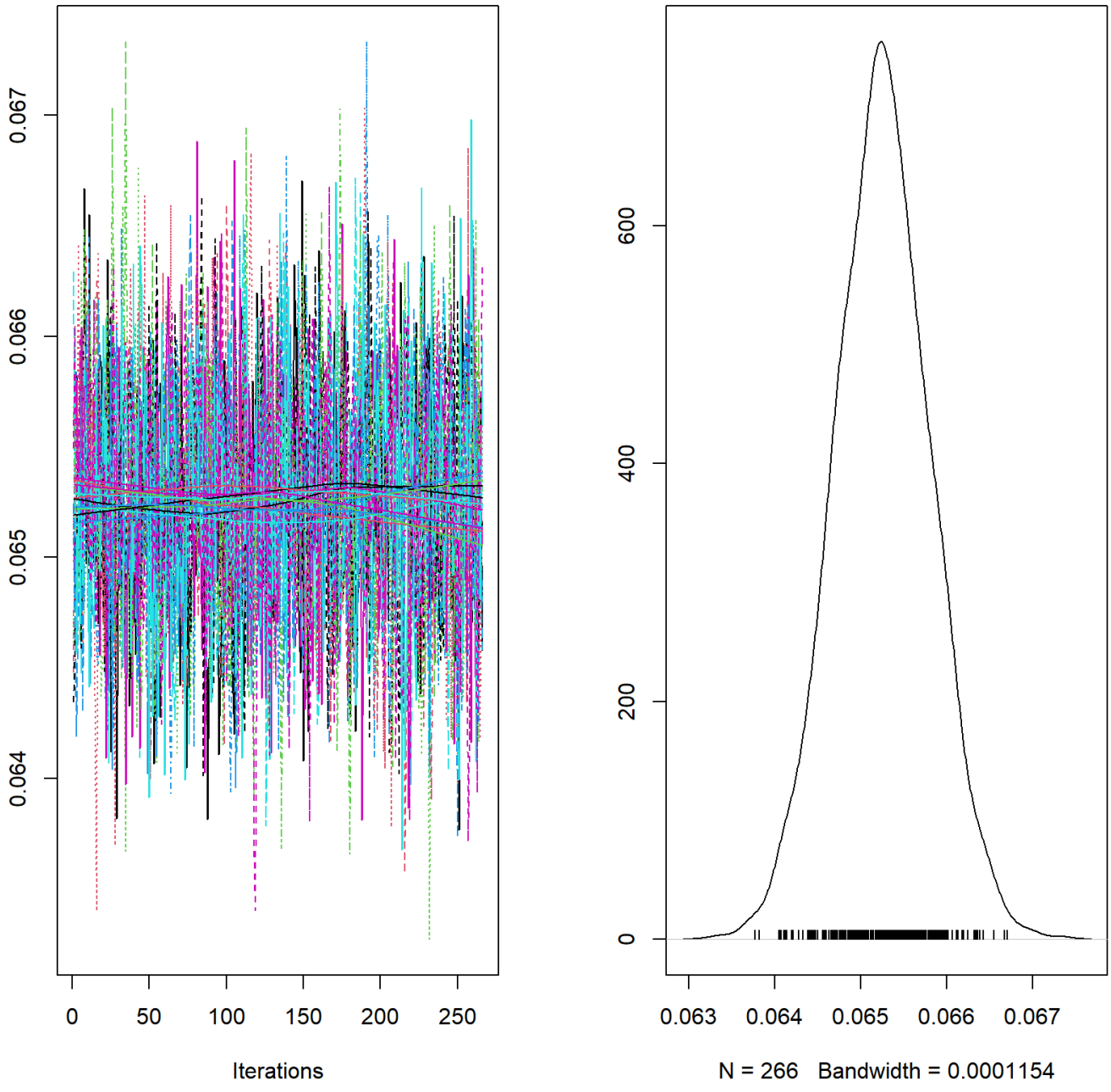
gelman.diag(mcmc_chains)
```

```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## [1,]          1       1.01
```

```
print(coda::effectiveSize(mcmc_chains))
```

```
##     var1
## 3402.778
```

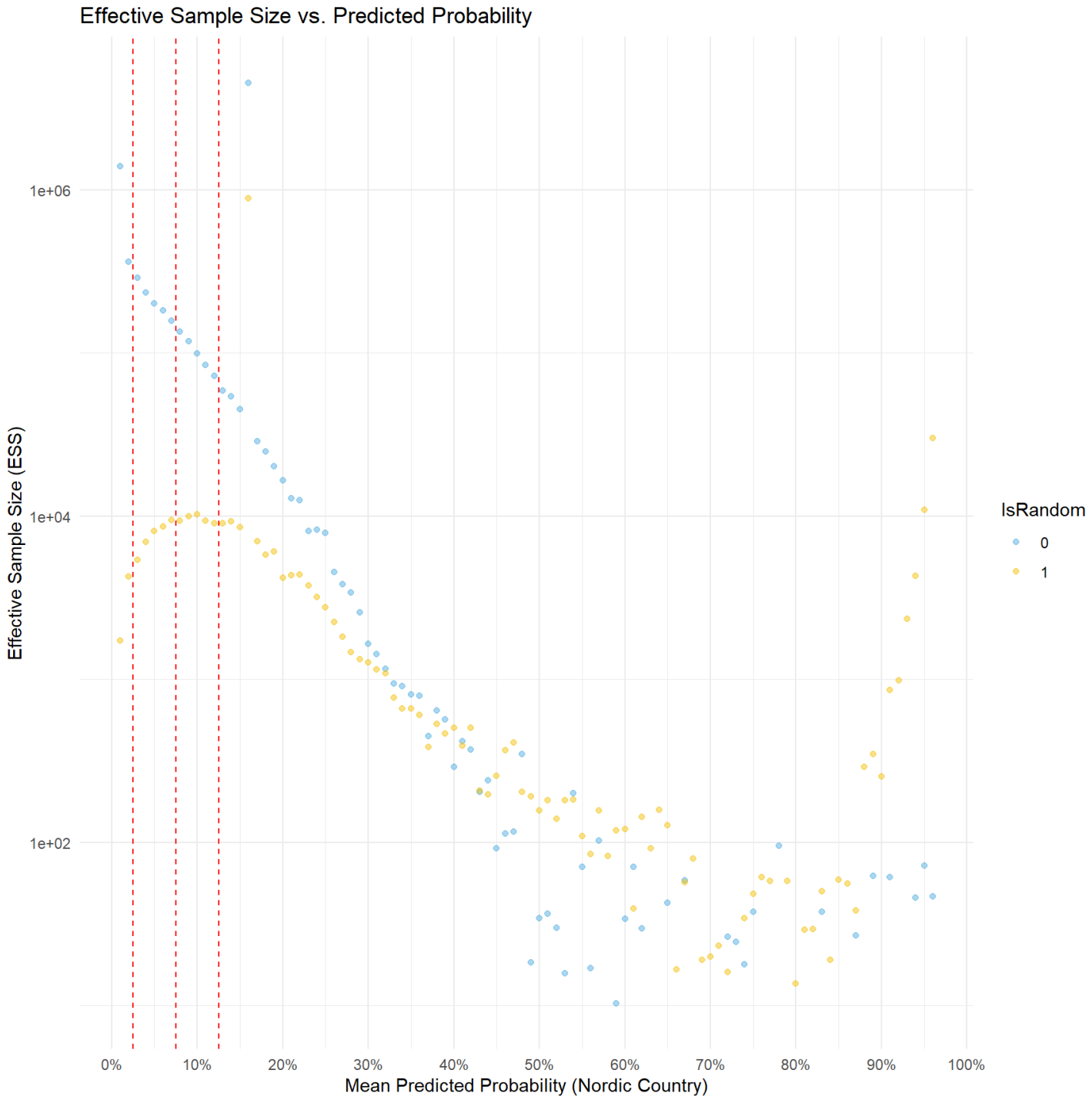
```
plot(mcmc_chains)
```



```
pred_prob_matrix <- modelRand2$yhat.train |> pnorm()
# Calculate ESS for each column (teacher) by looking across samples for each case
ess_values <- apply(pred_prob_matrix, 2, effectiveSize)
# Calculate the mean probability for each teacher
mean_probs <- colMeans(pred_prob_matrix)
# Create a data frame combining Teacher ID, Mean Probability, and ESS
teacher_diagnostics <- data.frame(
  IsRandom=modelRand2$fit$data@y,
  teacher_id = 1:length(mean_probs),
  mean_prob = mean_probs,
  ess = ess_values  )

cutoff_zone <- teacher_diagnostics[teacher_diagnostics$mean_prob >= YoudenCutoff$optimal[4]-0.05
&
                                   teacher_diagnostics$mean_prob <= YoudenCutoff$optimal[4]+0.05,
]

teacher_diagnostics |>
  mutate(mean_prob=cut(mean_prob,seq(0,1,0.01),include.lowest=T) |> factor() |> as.integer() ,
         mean_prob=mean_prob/100,
         IsRandom=factor(IsRandom) ) |>
  group_by(mean_prob,IsRandom) |>
  summarize(ess=sum(ess,na.rm=T) , .groups="drop") |>
  ggplot(aes(x = mean_prob, y = ess,color=IsRandom)) +
  geom_point(alpha = 0.5) +
  geom_vline(xintercept = c(max(YoudenCutoff$optimal[4]-0.05,0),
                            YoudenCutoff$optimal[4],
                            min(YoudenCutoff$optimal[4]+0.05,1)),
             linetype = "dashed", color = "red") +
  labs(title = "Effective Sample Size vs. Predicted Probability",
       x = "Mean Predicted Probability (Nordic Country)",
       y = "Effective Sample Size (ESS)") +
  theme_minimal()+
  scale_x_continuous(breaks=seq(0,1,0.1),label=scales::percent)+
  scale_y_log10()
```

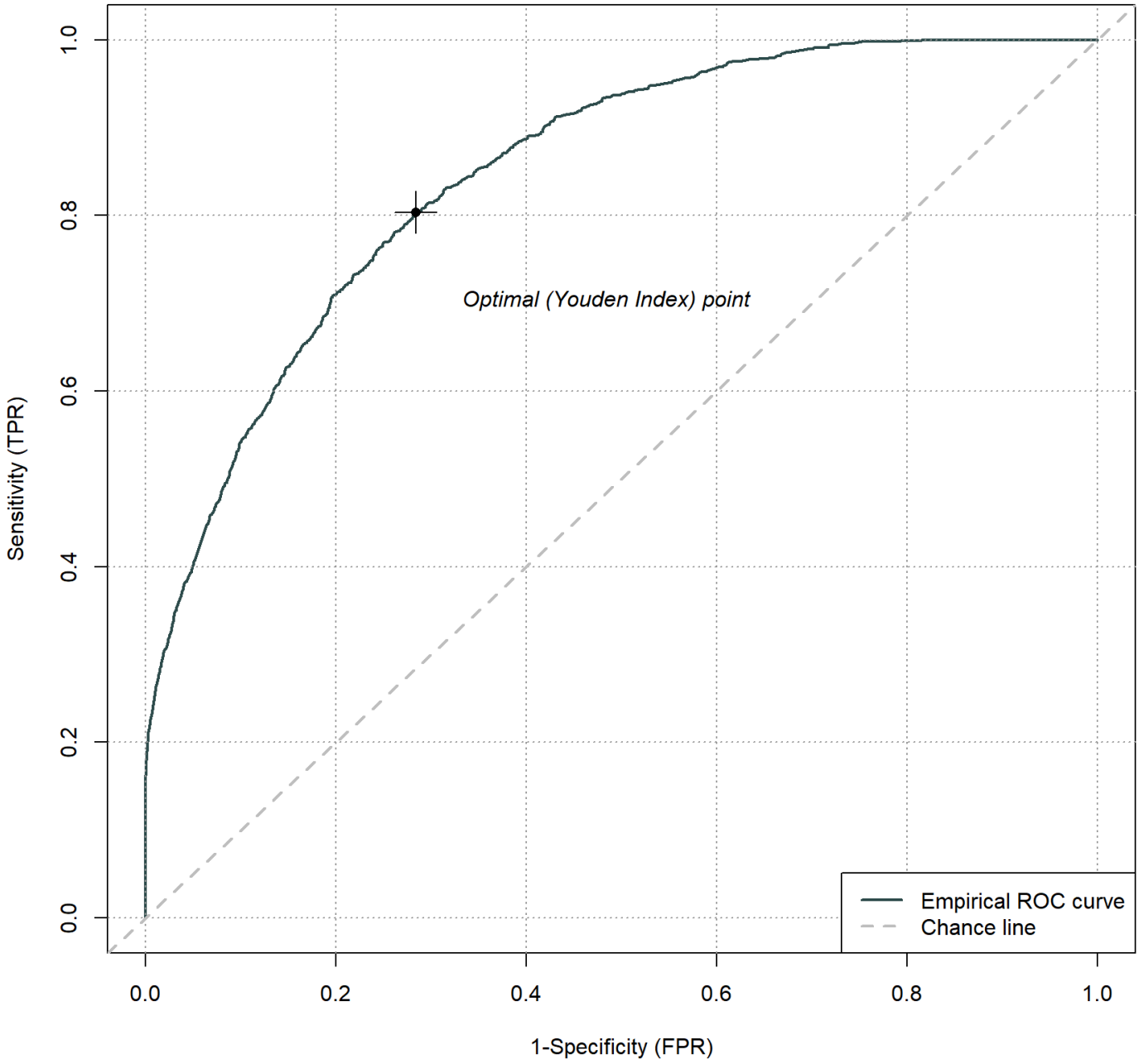


Results

```
str( YoudenCutoff )
```

```
## List of 6
##  $ method                    : chr "empirical"
##  $ AUC                       : num 0.861
##  $ Cutoff                    : num [1:156146] Inf 1 1 1 1 ...
##  $ TPR/Sensitivity/Recall    : num [1:156146] 0.00 9.66e-05 1.93e-04 2.90e-04 3.87e-04 ...
##  $ FPR/1-Specificity         : num [1:156146] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.5422 0.2503 0.7924 0.0751
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

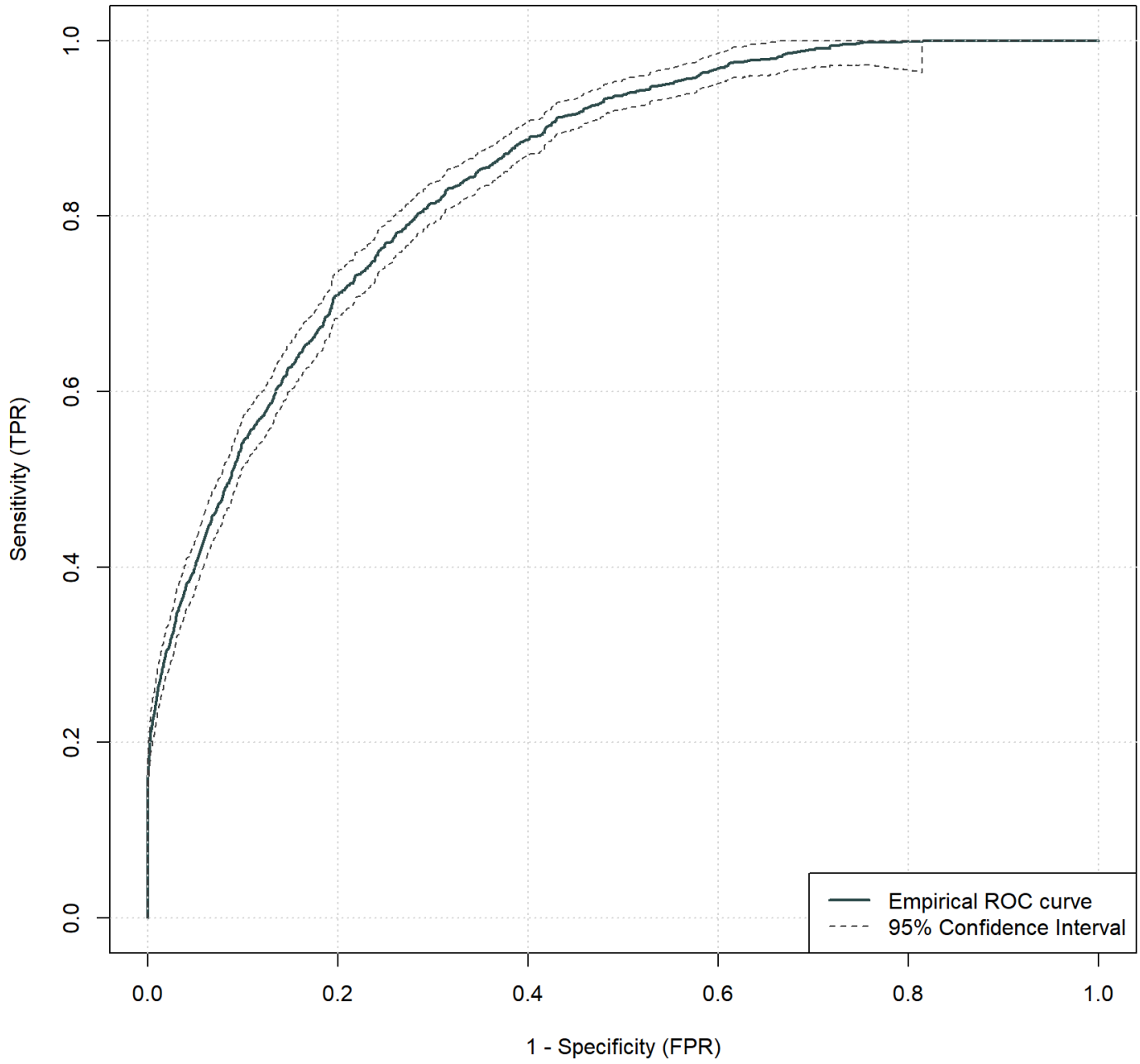
```
ROCit_obj_Random2 <- rocit(score=apply(modelRand2$yhat.test |> pnorm(),2,mean) ,class=Test$Random
2,negref =0)
str( ROCIT_Random2 <- plot(ROCit_obj_Random2) )
```




```
## List of 6
##  $ method                    : chr "empirical"
##  $ AUC                       : num 0.848
##  $ Cutoff                    : num [1:30079] Inf 1 1 1 1 ...
##  $ TPR/Sensitivity/Recall    : num [1:30079] 0 0.00085 0.0017 0.00255 0.0034 ...
##  $ FPR/1-Specificity         : num [1:30079] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.5198 0.2838 0.8036 0.0688
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

```
ciROC(ROCit_obj_Random2, method = "binomial") %>% plot(legend = TRUE)
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm): collapsing to unique
'x' values
```




```
ciAUC(ROCit_obj_Random2, delong = TRUE)
```

```
##
##    estimated AUC : 0.847969000306759
##    AUC estimation method : empirical
##
##    CI of AUC, delong method of variance used
##    confidence level = 95%
##    lower = 0.837649974044678     upper = 0.858288026568839
```

```
save(ROCIT_Random2,ROCit_obj_Random2,YoudenCutoff,file="ROCIT_Random2_Test.rData")
```

Tables for classification

```
Test %>%
  transmute(CNTRY,predicted=apply(modelRand2$yhat.test |> pnorm(),2,mean)>=YoudenCutoff$optimal
[4]) %>%
  table
```

```
##      predicted
## CNTRY FALSE TRUE
##   ALB   364   40
##   ARE   416   23
##   AUS    89  226
##   AUT   285  166
##   AZE   420   15
##   BEL   718  466
##   BGR   357   28
##   BHR   301   12
##   BRA   205  105
##   CAB    58  135
##   CHL    74  161
##   COL   228   88
##   CRI   202  106
##   CSH   366   47
##   CYP   125   62
##   CZE  1576  802
##   DNK   150  106
##   ESP  2785 2320
##   EST    77  265
##   FIN   236  213
##   FRA   186  106
##   HRV   194  122
##   HUN   323   82
##   ISL    67   84
##   ISR   278   56
##   ITA   290  101
##   JPN   277   89
##   KAZ  3256  111
##   KOR   234   80
##   LTU   296  137
##   LVA   230  180
##   MAR  2045   71
##   MKD   351   53
##   MLT   122  132
##   MNE   101   40
##   NLD    91   42
##   NOR    65   90
##   NZL    35   95
##   POL   251  129
##   PRT   200  169
##   ROU   348   47
##   SAU   290   44
##   SGP   290   50
##   SRB   314  101
##   SVK   296  131
##   SVN    26  258
##   SWE   121  185
##   TUR   895   59
##   USA    96  104
##   UZB   451    8
##   VNM   449    1
##   XKX   291   38
##   ZAF   259   84
```

```
Test %>%
  mutate(predicted = apply(modelRand2$yhat.test |> pnorm(),2,mean),
         Random2=factor(Random2,1:0,c("Random","Non-Random")) ) %>%
  group_by(CNTRY,Random2) %>%
  summarize(Unwg_pred=mean(predicted>=YoudenCutoff$optimal[4]),
            Wgt_pred =weighted.mean(predicted>=YoudenCutoff$optimal[4],unlabelled(TCHWGT)) ,
            .groups="drop") %>%
  ggplot(aes(y=CNTRY,color=Random2))+
    geom_point(aes(x=Unwg_pred,shape="Unweighted"))+
    geom_point(aes(x=Wgt_pred,shape="Weighted"))+
    labs(x="Percentage of teachers predicted to be Random",title="Test")
```

